

# The Sen API

## Interface Control Document

An example, generated from the data model by Sen. The reference that follows — every package, every type, every table — is written by the tool and is rewritten whenever the model changes.

This page is not. It is an ordinary Typst file included at a point the generated document leaves open, which is how a project puts its own cover, approvals and appendices around a reference it never has to edit.

Placeholder for your own project or product assets

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# About this document

This is an example published with Sen's documentation. It was produced by running the model through `sen generate typst` and compiling the result, with no editing in between.

Everything from *Model overview* onwards is generated. It follows the model and only the model: if a type is added, renamed or given a new description, the next build says so, and nothing anyone wrote by hand is lost in the process.

This page, and the cover before it, are not generated. They are Typst files of your own that the document includes at points it leaves open for you:

|                     |                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------|
| -- front-matter     | A cover, before the contents.                                                                               |
| -- before-reference | Anything between the contents and the model — a scope, a set of conventions, an approvals table. This page. |
| -- after-reference  | Appendices, after the model.                                                                                |
| -- style            | Every decision about how the document looks.                                                                |

The generator does not read any of those files. It writes an `#raw("#include")` and Typst resolves it when the document is compiled, so a cover page can be anything Typst can set — and changing it never means regenerating the model.

# Model overview

314 types in 13 packages. 13 classes in 11 independent hierarchies.

## Packages

| Package    | Types | Object classes | Fixed records | Enumerations | Variant records | Arrays | Optionals |
|------------|-------|----------------|---------------|--------------|-----------------|--------|-----------|
| components | —     | —              | —             | —            | —               | —      | —         |
| ether      | 31    | —              | 20            | 1            | 3               | 4      | 3         |
| explorer   | 1     | —              | 1             | —            | —               | —      | —         |
| influx     | 3     | —              | 1             | 1            | —               | 1      | —         |
| jsonrpc    | 41    | 2              | 17            | —            | 1               | 8      | 13        |
| logmaster  | 3     | 2              | 1             | —            | —               | —      | —         |
| py         | 3     | 1              | 1             | —            | —               | 1      | —         |
| recorder   | 6     | 1              | 2             | 1            | —               | 2      | —         |
| replayer   | 4     | 2              | 1             | 1            | —               | —      | —         |
| rest       | 37    | —              | 20            | 5            | —               | 9      | 3         |
| shell      | 27    | 1              | 20            | 3            | 1               | 2      | —         |
| db         | 3     | —              | 3             | —            | —               | —      | —         |
| kernel     | 139   | 4              | 70            | 20           | 7               | 32     | 6         |
| log        | 16    | —              | 11            | 1            | 1               | 3      | —         |

## Class hierarchy across all packages

|                                                     |    |
|-----------------------------------------------------|----|
| <a href="#">components.jsonrpc.JsonRpcServer</a>    | 15 |
| <a href="#">components.jsonrpc.StaticFileServer</a> | 18 |
| <a href="#">components.logmaster.LogMaster</a>      | 26 |
| <a href="#">components.logmaster.Logger</a>         | 26 |
| <a href="#">components.py.PythonInterpreter</a>     | 27 |
| <a href="#">components.recorder.Recorder</a>        | 28 |
| <a href="#">components.replayer.Replay</a>          | 30 |
| <a href="#">components.replayer.Replayer</a>        | 30 |
| <a href="#">components.shell.Shell</a>              | 38 |
| <a href="#">kernel.KernelApi</a>                    | 44 |
| <a href="#">kernel.VirtualClock</a>                 | 44 |
| └─ <a href="#">kernel.VirtualKernelClock</a>        | 44 |
| └─ <a href="#">kernel.VirtualMasterClock</a>        | 44 |

# Reading the tables

Every property carries three flags, in this order.

| Flag | Meaning                                                                              |
|------|--------------------------------------------------------------------------------------|
| R0   | Read-only: the value is published by whoever owns it and cannot be set from outside. |
| RW   | Read-write: the value can be set as well as read.                                    |
| D    | Dynamic: the value changes over the life of an instance.                             |
| S    | Static: the value is fixed once an instance exists.                                  |
| C    | Confirmed: delivery is acknowledged, and the update is retried until it arrives.     |
| BE   | Best effort: the update is sent once and may be lost.                                |

# components

# ether

31 types.

## Fixed records

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                          |    |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">BusConfig</a>             | Remember to set these to the same value in all related Sen instances                                                                                                                                                                                                                                                                                                                                                     | 7  |
| <a href="#">BusJoined</a>             |                                                                                                                                                                                                                                                                                                                                                                                                                          | 7  |
| <a href="#">BusLeft</a>               |                                                                                                                                                                                                                                                                                                                                                                                                                          | 7  |
| <a href="#">ByteRange</a>             | Range of values of a u8                                                                                                                                                                                                                                                                                                                                                                                                  | 8  |
| <a href="#">Configuration</a>         | Overall component configuration                                                                                                                                                                                                                                                                                                                                                                                          | 8  |
| <a href="#">DiscoveryHubAddress</a>   | TCP-based discovery hub parameters                                                                                                                                                                                                                                                                                                                                                                                       | 8  |
| <a href="#">Endpoint</a>              |                                                                                                                                                                                                                                                                                                                                                                                                                          | 8  |
| <a href="#">Ephemeral</a>             | Let the operating system select an ephemeral port                                                                                                                                                                                                                                                                                                                                                                        | 8  |
| <a href="#">Hello</a>                 |                                                                                                                                                                                                                                                                                                                                                                                                                          | 8  |
| <a href="#">MulticastAddressRange</a> | Inclusive IPv4 multicast address range                                                                                                                                                                                                                                                                                                                                                                                   | 9  |
| <a href="#">MulticastDiscovery</a>    | Multicast-based discovery parameters                                                                                                                                                                                                                                                                                                                                                                                     | 9  |
| <a href="#">PinnedPort</a>            | Bind exactly this port, fail closed if taken                                                                                                                                                                                                                                                                                                                                                                             | 9  |
| <a href="#">PortConfig</a>            | This configuration is per process and does not need to match across Sen instances. Exclusions apply to a pinned port, not to an ephemeral one: an ephemeral bind asks the OS for any free port and does not learn which it was given. udpUnicast cannot be pinned. Its socket is per peer and UDP has no four-tuple to demultiplex on, so a second peer either fails to bind or silently takes the first peer's traffic. | 9  |
| <a href="#">PortRange</a>             | Inclusive port range                                                                                                                                                                                                                                                                                                                                                                                                     | 9  |
| <a href="#">ProbePortRange</a>        | Bind within [min,max], random start, walk forward                                                                                                                                                                                                                                                                                                                                                                        | 10 |
| <a href="#">ProtocolVersion</a>       |                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 |
| <a href="#">QueueConfig</a>           |                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 |
| <a href="#">Ready</a>                 |                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 |
| <a href="#">SessionPresenceBeam</a>   |                                                                                                                                                                                                                                                                                                                                                                                                                          | 10 |
| <a href="#">TcpDiscovery</a>          | TCP-based discovery parameters                                                                                                                                                                                                                                                                                                                                                                                           | 10 |

## BusConfig STRUCTURE

Remember to set these to the same value in all related Sen instances

| Name                | Type                                        | Description                                                                                                                                                                 |
|---------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| multicastRange      | <a href="#">MulticastRange</a>              | Defaults to the Organization Local scope (RFC 2365): [239, [192-195], [0-255], [0-255] Remember to keep a range wide enough for avoiding collisions if you have many buses. |
| multicastExclusions | <a href="#">MulticastAddress Exclusions</a> | Address ranges that must not be allocated                                                                                                                                   |
| multicastPort       | <a href="#">u16</a>                         | Defaults to 50985 Change this in case you have limited control over the infrastructure.                                                                                     |
| multicastDisabled   | <a href="#">bool</a>                        | If true, no multicast traffic is performed and all the bus settings above become meaningless. Defaults to false.                                                            |

Named by [Configuration](#).

## BusJoined STRUCTURE

| Name          | Type                   | Description |
|---------------|------------------------|-------------|
| participantId | <a href="#">u32</a>    |             |
| busId         | <a href="#">u32</a>    |             |
| busName       | <a href="#">string</a> |             |

Named by [ControlMessage](#).

## BusLeft STRUCTURE

| Name          | Type                | Description |
|---------------|---------------------|-------------|
| participantId | <a href="#">u32</a> |             |
| busId         | <a href="#">u32</a> |             |

| Name    | Type                   | Description |
|---------|------------------------|-------------|
| busName | <a href="#">string</a> |             |

Named by [ControlMessage](#).

## ByteRange STRUCTURE

Range of values of a u8

| Name | Type               | Description        |
|------|--------------------|--------------------|
| min  | <a href="#">u8</a> | minimum range      |
| max  | <a href="#">u8</a> | needs to be >= min |

Named by [MulticastRange](#).

## Configuration STRUCTURE

Overall component configuration

| Name               | Type                            | Description                                        |
|--------------------|---------------------------------|----------------------------------------------------|
| discovery          | <a href="#">DiscoveryConfig</a> | Discovery-related parameters                       |
| busOutQueue        | <a href="#">QueueConfig</a>     | Queue for broadcasting to buses                    |
| processUdpOutQueue | <a href="#">QueueConfig</a>     | Queue for communicating with processes over UDP    |
| processTcpOutQueue | <a href="#">QueueConfig</a>     | Queue for communicating with processes over TCP    |
| busConfig          | <a href="#">BusConfig</a>       | Bus transport configuration                        |
| portExclusions     | <a href="#">PortExclusions</a>  | Port ranges that this process must not use         |
| portConfig         | <a href="#">PortConfig</a>      | If not present, all port bindings are Ephemeral    |
| runDiscoveryHub    | <a href="#">u16</a>             | If present, run the TCP discovery hub on this port |
| networkDevice      | <a href="#">string</a>          | If present, the network device to use              |

## DiscoveryHubAddress STRUCTURE

TCP-based discovery hub parameters

| Name | Type                   | Description                                               |
|------|------------------------|-----------------------------------------------------------|
| port | <a href="#">u16</a>    | Port where the discovery hub is running                   |
| host | <a href="#">string</a> | name or IP of the host where the discovery hub is running |

Named by [TcpDiscovery](#).

## Endpoint STRUCTURE

| Name | Type                | Description |
|------|---------------------|-------------|
| ip   | <a href="#">u32</a> |             |
| port | <a href="#">u16</a> |             |

Named by [EndpointList](#).

## Ephemeral STRUCTURE

Let the operating system select an ephemeral port

Named by [PortBinding](#).

## Hello STRUCTURE

| Name | Type                                   | Description |
|------|----------------------------------------|-------------|
| info | <a href="#">sen.kernel.ProcessInfo</a> |             |

| Name    | Type                            | Description |
|---------|---------------------------------|-------------|
| udpPort | <a href="#">u16</a>             |             |
| version | <a href="#">ProtocolVersion</a> |             |

Named by [ControlMessage](#).

## MulticastAddressRange STRUCTURE

Inclusive IPv4 multicast address range

| Name | Type                   | Description |
|------|------------------------|-------------|
| min  | <a href="#">string</a> |             |
| max  | <a href="#">string</a> |             |

Named by [MulticastAddressExclusions](#).

## MulticastDiscovery STRUCTURE

Multicast-based discovery parameters

| Name                       | Type                         | Description                                                  |
|----------------------------|------------------------------|--------------------------------------------------------------|
| beamPeriod                 | <a href="#">sen.Duration</a> | How often are we beaming our presence                        |
| beamExpirationTime         | <a href="#">sen.Duration</a> | Expiry time used by the beam tracker                         |
| mcastGroup                 | <a href="#">string</a>       | Defaults to '239.255.0.44'                                   |
| port                       | <a href="#">u16</a>          | Defaults to 60543                                            |
| device                     | <a href="#">string</a>       | Device to use for the network communication                  |
| allowVirtual<br>Interfaces | <a href="#">bool</a>         | If true, virtual (NO_CARRIER) interfaces are also considered |

Named by [DiscoveryConfig](#).

## PinnedPort STRUCTURE

Bind exactly this port, fail closed if taken

| Name | Type                | Description |
|------|---------------------|-------------|
| port | <a href="#">u16</a> |             |

Named by [PortBinding](#).

## PortConfig STRUCTURE

This configuration is per process and does not need to match across Sen instances. Exclusions apply to a pinned port, not to an ephemeral one: an ephemeral bind asks the OS for any free port and does not learn which it was given. udpUnicast cannot be pinned. Its socket is per peer and UDP has no four-tuple to demultiplex on, so a second peer either fails to bind or silently takes the first peer's traffic.

| Name        | Type                        | Description                             |
|-------------|-----------------------------|-----------------------------------------|
| tcpAcceptor | <a href="#">PortBinding</a> | inbound TCP listener                    |
| udpUnicast  | <a href="#">PortBinding</a> | per-peer UDP unicast; must be Ephemeral |
| tcpSource   | <a href="#">PortBinding</a> | outbound TCP source port                |

Named by [Configuration](#).

## PortRange STRUCTURE

Inclusive port range

| Name | Type                | Description |
|------|---------------------|-------------|
| min  | <a href="#">u16</a> |             |

| Name | Type | Description |
|------|------|-------------|
| max  | u16  |             |

Named by [PortExclusions](#).

## ProbePortRange STRUCTURE

Bind within [min,max], random start, walk forward

| Name | Type | Description |
|------|------|-------------|
| min  | u16  |             |
| max  | u16  |             |

Named by [PortBinding](#).

## ProtocolVersion STRUCTURE

| Name   | Type | Description |
|--------|------|-------------|
| kernel | u32  |             |
| ether  | u32  |             |

Named by [Hello](#).

## QueueConfig STRUCTURE

| Name           | Type                                | Description                                              |
|----------------|-------------------------------------|----------------------------------------------------------|
| evictionPolicy | <a href="#">QueueEvictionPolicy</a> | What to do when a queue is full                          |
| maxSize        | u64                                 | Maximum element count, 0 means unbounded                 |
| warningLevel   | u64                                 | Element count where warning will be issued, 0 means none |

Named by [Configuration](#).

## Ready STRUCTURE

Named by [ControlMessage](#).

## SessionPresenceBeam STRUCTURE

| Name            | Type                                   | Description |
|-----------------|----------------------------------------|-------------|
| protocolVersion | u16                                    |             |
| info            | <a href="#">sen.kernel.ProcessInfo</a> |             |
| beamPeriod      | <a href="#">sen.Duration</a>           |             |
| endpoints       | <a href="#">EnpointList</a>            |             |

## TcpDiscovery STRUCTURE

TCP-based discovery parameters

| Name                       | Type                                | Description                                                  |
|----------------------------|-------------------------------------|--------------------------------------------------------------|
| beamPeriod                 | <a href="#">sen.Duration</a>        | How often are we beaming our presence                        |
| beamExpirationTime         | <a href="#">sen.Duration</a>        | Expiry time used by the beam tracker                         |
| hubAddress                 | <a href="#">DiscoveryHubAddress</a> | Where to find the discovery hub                              |
| allowVirtual<br>Interfaces | bool                                | If true, virtual (NO_CARRIER) interfaces are also considered |

Named by [DiscoveryConfig](#).

Enumerations

QueueEvictionPolicy

What to do when a queue is full

11

QueueEvictionPolicy

ENUMERATION

What to do when a queue is full

Held as u32.

|            |   |
|------------|---|
| dropOldest | 0 |
| dropNewest | 1 |

Named by QueueConfig.

Variant records

ControlMessage

DiscoveryConfig

PortBinding

Configuration parameters related to the discovery protocol

11

11

11

ControlMessage

VARIANT

| Type      | Description |
|-----------|-------------|
| Hello     |             |
| Ready     |             |
| BusJoined |             |
| BusLeft   |             |

DiscoveryConfig

VARIANT

Configuration parameters related to the discovery protocol

| Type               | Description |
|--------------------|-------------|
| MulticastDiscovery |             |
| TcpDiscovery       |             |

Named by Configuration.

PortBinding

VARIANT

| Type           | Description |
|----------------|-------------|
| Ephemeral      |             |
| PinnedPort     |             |
| ProbePortRange |             |

Named by PortConfig.

Arrays

EnpointList

MulticastAddressExclusions

MulticastRange

PortExclusions

Range of values of a IPv4 multicast address

11

12

12

12

EnpointList

SEQUENCE

sequence<Endpoint, 27> EnpointList;

Named by SessionPresenceBeam.

## MulticastAddressExclusions SEQUENCE

```
sequence<MulticastAddressRange> MulticastAddressExclusions;
```

Named by [BusConfig](#).

## MulticastRange SEQUENCE

Range of values of a IPv4 multicast address

```
array<ByteRange, 4> MulticastRange;
```

Named by [BusConfig](#).

## PortExclusions SEQUENCE

```
sequence<PortRange> PortExclusions;
```

Named by [Configuration](#).

### Optionals

|                                       |    |
|---------------------------------------|----|
| <a href="#">MaybeDeviceName</a>       | 12 |
| <a href="#">MaybeDiscoveryHubPort</a> | 12 |
| <a href="#">MaybePortConfig</a>       | 12 |

## MaybeDeviceName OPTIONAL

```
optional<string> MaybeDeviceName;
```

## MaybeDiscoveryHubPort OPTIONAL

```
optional<u16> MaybeDiscoveryHubPort;
```

## MaybePortConfig OPTIONAL

```
optional<PortConfig> MaybePortConfig;
```

# explorer

1 types.

**Fixed records**

|                               |                                 |    |
|-------------------------------|---------------------------------|----|
| <a href="#">Configuration</a> | Overall component configuration | 13 |
|-------------------------------|---------------------------------|----|

## Configuration STRUCTURE

Overall component configuration

| Name       | Type                   | Description                           |
|------------|------------------------|---------------------------------------|
| layoutFile | <a href="#">string</a> | the imgui layout file. Defaults to "" |

# influx

3 types.

## Fixed records

[Configuration](#) Overall component configuration

14

## Configuration STRUCTURE

Overall component configuration

| Name            | Type                          | Description                                           |
|-----------------|-------------------------------|-------------------------------------------------------|
| samplingPeriod  | <a href="#">sen.Duration</a>  | sampling period for the recordings                    |
| selections      | <a href="#">SelectionList</a> | objects to be monitored (using sql)                   |
| telegrafAddress | <a href="#">string</a>        | ip or name of the Telegraf server                     |
| telegrafPort    | <a href="#">u16</a>           | port used by the Telegraf server                      |
| protocol        | <a href="#">Protocol</a>      | protocol used                                         |
| batchSize       | <a href="#">u32</a>           | points to batch before emission (0 means no batching) |

## Enumerations

[Protocol](#)

14

## Protocol ENUMERATION

Held as u8.

|     |   |
|-----|---|
| udp | 0 |
| tcp | 1 |

Named by [Configuration](#).

## Arrays

[SelectionList](#)

14

## SelectionList SEQUENCE

```
sequence<string> SelectionList;
```

Named by [Configuration](#).

# jsonrpc

41 types.

## Object classes

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">JsonRpcServer</a>    | JsonRpcServer -- the JSON-RPC over WebSocket external API, expressed as a Sen class. One instance per connected client lives on <code>`local.jsonrpc`</code> . Methods are the RPC surface; events are the server-pushed notifications. Wire framing (envelope, id, error mapping) is the dispatcher's job, not this class's. Wire-shape convention: JSON-RPC named params ( <code>"params": { ... }</code> ) use the same field names as the STL method's argument list. The dispatcher treats every method's params object as <code>{ &lt;argName&gt;: &lt;value&gt;, ... }</code> and walks <code>`method-&gt;getArgs()`</code> to build the VarList in declaration order. That lets the inbound dispatch path go through Sen's meta machinery with no per-method translation. | 15 |
| <a href="#">StaticFileServer</a> | Server-wide registry of in-memory static-file bundles, served alongside the WebSocket+JSON-RPC surface by the same uWS app. Internal-only: lives on the configurable control bus (defaults to <code>`local.jsonrpc_control`</code> ), not on the data-plane bus that hosts authenticated per-connection <code>JsonRpcServer</code> instances. The component publishes exactly one instance.                                                                                                                                                                                                                                                                                                                                                                                       | 18 |

## JsonRpcServer CLASS

JsonRpcServer -- the JSON-RPC over WebSocket external API, expressed as a Sen class. One instance per connected client lives on ``local.jsonrpc``. Methods are the RPC surface; events are the server-pushed notifications. Wire framing (envelope, id, error mapping) is the dispatcher's job, not this class's. Wire-shape convention: JSON-RPC named params (`"params": { ... }`) use the same field names as the STL method's argument list. The dispatcher treats every method's params object as `{ <argName>: <value>, ... }` and walks ``method->getArgs()`` to build the VarList in declaration order. That lets the inbound dispatch path go through Sen's meta machinery with no per-method translation.

| Name     | Type                     | Flags   | Description                       |
|----------|--------------------------|---------|-----------------------------------|
| identity | <a href="#">Identity</a> | RO S BE | Identity of the connected client. |

  

| Method              | Signature                                  | Description                                                                                                                                                                                                                                                                                                                              |
|---------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ping                | () → <a href="#">string</a>                | Liveness probe.                                                                                                                                                                                                                                                                                                                          |
| listTopology        | () → <a href="#">SessionInfoList</a>       | Snapshot of the currently-detected sessions, each with the buses currently detected inside it. The kernel's discovery surface is asynchronous; this returns whatever was visible at the moment of the call. Use <code>`subscribeTopology`</code> for live updates.                                                                       |
| subscribeTopology   | ()                                         | Sticky subscription to topology changes. Idempotent per connection. Wires a <code>`topologyChanged`</code> push immediately with the current full snapshot, then again on every session or bus add/remove. Cancel with <code>`unsubscribeTopology`</code> ; the subscription is also torn down automatically when the connection closes. |
| unsubscribeTopology | ()                                         | Cancel the topology subscription created by <code>`subscribeTopology`</code> . Idempotent.                                                                                                                                                                                                                                               |
| getTypes            | () → <a href="#">sen.kernel.StringList</a> | Returns the qualified names of every custom type registered with the kernel. Catalog only -- pair with <code>`getType`</code> to fetch a chosen spec. The bundled-with-objects path on <code>`interestUpdate.types`</code> remains the primary way clients learn about types they observe.                                               |

| Method               | Signature                                                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| getType              | (qualifiedName: <code>string</code> , withSchema: <code>bool</code> ) → <code>TypeLookupResult</code>                                                                     | Look up one type by qualified name. Returns the <code>CustomTypeSpec</code> ; when <code>`withSchema`</code> is true, also returns the JSON-Schema fragment for the type (empty string otherwise). Dependency chasing is the client's job: each schema fragment carries <code>`\$ref`</code> strings keyed by qualified name, resolved against whatever schema cache the client maintains.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| createInterest       | (interestName: <code>string</code> , query: <code>string</code> , subscribe: <code>SubscribeBlock</code> , withSchemas: <code>bool</code> )                               | Register a sticky interest defined by an STL query. Optional <code>`subscribe`</code> block pre-subscribes to properties/events on every matched object, including ones that arrive later. When <code>`withSchemas`</code> is true, every subsequent <code>`interestUpdate`</code> for this interest carries the JSON-Schema fragment for each previously-unseen type alongside the existing <code>`CustomTypeSpec`</code> . Re-registering the same name = <code>invalidParams</code> . Wire-ordering contract: the initial <code>`interestUpdate.added`</code> for the current match set is emitted <i>before</i> this RPC's response lands on the wire. Clients must accept pushes for an interest whose response has not yet been observed.                                                                                                                                                                                                    |
| releaseInterest      | (interestName: <code>string</code> )                                                                                                                                      | Drop an interest and every subscription living under it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| listObjects          | (interestName: <code>string</code> ) → <code>ObjectInfos</code>                                                                                                           | List the names and classes of objects currently in an interest's match set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| getProperty          | (interestName: <code>string</code> , objectName: <code>string</code> , propertyName: <code>string</code> ) → <code>string</code>                                          | Read a property's current value, synchronously.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| getObjectsBatchState | (interestName: <code>string</code> , objectNames: <code>sen.kernel.StringList</code> , propertyNames: <code>sen.kernel.StringList</code> ) → <code>ObjectStateList</code> | Batched per-object snapshot. Walks the interest's match set in one pass instead of forcing the caller to issue N*M <code>`getProperty`</code> round-trips. Omitting <code>`objectNames`</code> reads every object currently matched; omitting <code>`propertyNames`</code> reads every property of each object (the class hierarchy is walked, so inherited members are included). Names from <code>`objectNames`</code> that aren't currently in the match set are silently skipped -- the caller compares its requested list against the returned <code>`objectName`</code> 's to detect omissions. Names from <code>`propertyNames`</code> that don't exist on a given object's class are recorded per-object under <code>`errors`</code> , so a heterogeneous batch still returns useful partial results. Bounded: the call rejects with <code>`invalidParams`</code> when the resolved object set exceeds 512 entries (either via an explicit |

| Method              | Signature                                                                                                                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| setProperty         | (interestName: <code>string</code> , objectName: <code>string</code> , propertyName: <code>string</code> , value: <code>string</code> )                        | <p><code>`objectNames`</code> list of that length, or via an implicit walk over a match set that has grown that large). Page the request by passing <code>`objectNames`</code>.</p> <p>Write a property's value (sugar over invoking the generated setter). Hand-routed in</p> <p><code>`Dispatcher::handleSetProperty`</code>: the dispatcher issues <code>`invokeUntyped`</code> against the property's setter and wires the response through the kernel callback in a single tick. The typed <code>`setPropertyImpl`</code> override in the C++ <code>`Server`</code> exists only to satisfy the codegen-generated abstract base and is unreachable at runtime. See <code>`architecture.md`</code> "Method dispatch" for why this method (and <code>`invoke`</code>) bypass the generic meta path.</p> |
| subscribeProperty   | (interestName: <code>string</code> , objectName: <code>string</code> , propertyName: <code>string</code> , maxRateHz: <code>f64</code> )                       | <p>Subscribe to <code>propertyChanged</code> notifications for one (interest, object, property). Sticky: persists across object churn; auto-rewires when the named object (re)enters the match set. Snapshot of the current value is delivered immediately after wiring. The throttle is per-object, shared across every property/event subscribed for the same object. An explicit <code>`maxRateHz`</code> overrides the current rate for the object; omitting it preserves whatever rate is already in effect (so adding a per-property sub doesn't accidentally retune existing subs back to unlimited).</p>                                                                                                                                                                                          |
| unsubscribeProperty | (interestName: <code>string</code> , objectName: <code>string</code> , propertyName: <code>string</code> )                                                     | <p>Cancel a property subscription previously created with <code>subscribeProperty</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| subscribeEvent      | (interestName: <code>string</code> , objectName: <code>string</code> , eventName: <code>string</code> )                                                        | <p>Subscribe to <code>eventTriggered</code> notifications for one (interest, object, event).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| unsubscribeEvent    | (interestName: <code>string</code> , objectName: <code>string</code> , eventName: <code>string</code> )                                                        | <p>Cancel an event subscription previously created with <code>subscribeEvent</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| subscribeAll        | (interestName: <code>string</code> , objectName: <code>string</code> , maxRateHz: <code>f64</code> )                                                           | <p>Subscribe to every property and every event of one named object. Wildcard expansion against the object's actual class (including parent classes). Object must currently be in the match set; absent objects are <code>invalidParams</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| unsubscribeAll      | (interestName: <code>string</code> , objectName: <code>string</code> )                                                                                         | <p>Cancel an all-members subscription previously created with <code>subscribeAll</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| invoke              | (interestName: <code>string</code> , objectName: <code>string</code> , methodName: <code>string</code> , argsJson: <code>string</code> ) → <code>string</code> | <p>Call a method on a kernel object and await its return value. Hand-routed in <code>`Dispatcher::handleInvoke`</code>: validates the args, issues <code>`invokeUntyped`</code>, and wires the response through the kernel callback in a single tick. The typed</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Method | Signature | Description                                                                                                                                                                                                                     |
|--------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |           | <code>`invokeImpl`</code> override in the C++<br><code>`Server`</code> exists only to satisfy the<br>codegen-generated abstract base and<br>is unreachable at runtime. See<br><code>`architecture.md`</code> "Method dispatch". |

  

| Event                | Payload                                                                                                                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| propertyChanged      | (interestName: <code>string</code> , objectName: <code>string</code> , values: <code>PropertyValueList</code> , timestamp: <code>sen.TimeStamp</code> )                                                         | Bundled per-object snapshot of changed properties. Per-object bundling, not per-property: any property change on one object lands in a single envelope.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| eventTriggered       | (interestName: <code>string</code> , objectName: <code>string</code> , eventName: <code>string</code> , args: <code>string</code> , timestamp: <code>sen.TimeStamp</code> )                                     | An event fired on one object in one interest. The event signature is decided by the source object's class, not by JSON-RPC.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| interestUpdate       | (interestName: <code>string</code> , added: <code>AddedObjectList</code> , removed: <code>sen.kernel.StringList</code> , types: <code>sen.kernel.CustomTypeSpecList</code> , typeSchemas: <code>string</code> ) | Match-set delta for one interest. The dispatcher tracks knownTypes / knownSchemas per connection and only re-sends specs / schemas new to the client.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| topologyChanged      | (sessions: <code>SessionInfoList</code> )                                                                                                                                                                       | Full snapshot of the topology after a session or bus add/remove. Pushed once on <code>`subscribeTopology`</code> (with the current state) and again on every change. Snapshot form rather than deltas: topology changes are rare, and the snapshot keeps client logic trivial and idempotent.                                                                                                                                                                                                                                                                                                                       |
| notificationsDropped | (count: <code>u64</code> )                                                                                                                                                                                      | Sent once when outbound backpressure clears on this connection, reporting how many unreliable notifications were dropped while the buffer was above the high watermark. Emitted reliably on purpose: the notice about dropped messages must not be droppable. A client that receives this has an incomplete picture and cannot tell which updates it missed -- only that it missed some. The correct response is to re-read what it displays ( <code>getObjectsBatchState</code> over the affected interests) rather than to wait for the next change, which may never come for a property that has stopped moving. |

## StaticFileServer CLASS

Server-wide registry of in-memory static-file bundles, served alongside the WebSocket+JSON-RPC surface by the same uWS app. Internal-only: lives on the configurable control bus (defaults to ``local.jsonrpc_control``), not on the data-plane bus that hosts authenticated per-connection `JsonRpcServer` instances. The component publishes exactly one instance.

| Name              | Type                                  | Flags  | Description                                                                                                                                                                         |
|-------------------|---------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| registeredBundles | <code>RegisteredBundleInfoList</code> | RO D C | Currently-registered bundles. Externally read-only; the component mutates this whenever a bundle is registered or unregistered. Subscribe to receive notifications on every change. |

  

| Method               | Signature                                               | Description                                                                                    |
|----------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------|
| registerStaticBundle | (bundle: <code>StaticBundle</code> ) → <code>u64</code> | Register a bundle. The bundle is copied into the registry; the caller may release the argument |

| Method                 | Signature                        | Description                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| unregisterStaticBundle | (bundleId: <a href="#">u64</a> ) | afterwards. Routes are wired up asynchronously on the uWS thread; the returned bundleId is stable and may be used immediately with <code>unregisterStaticBundle</code> .<br><br>Stop serving the bundle. The slot in the registry is nulled; the underlying uWS route remains registered (uWS does not expose a clean route-removal API) but becomes inert and returns 404. No-op if <code>bundleId</code> is unknown. |

## Fixed records

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">AddedObjectEntry</a>     | One object newly entering an interest's match set.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 19 |
| <a href="#">Configuration</a>        | JSON-RPC over WebSocket component configuration. Only <code>address</code> and <code>port</code> are required; every other field falls back to a documented default.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 20 |
| <a href="#">ConnectionLimits</a>     | Per-connection buffer / timeout knobs consulted on the uWS thread for the lifetime of the server. Every field is optional; unset fields fall back to the defaults resolved inside <code>ws_server.cpp</code> at the use site.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 20 |
| <a href="#">Identity</a>             | Identity of the connected client. Set once on connect by the dispatcher from the WebSocket upgrade authenticator. Read-only externally. Other components / admin tools can <code>SELECT * FROM local.jsonrpc</code> to enumerate currently-connected clients.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 20 |
| <a href="#">NamedSelection</a>       | Named arm of <code>MemberSelector</code> : matches only the listed members. Wire shape: <code>{type: "NamedSelection", value: {memberNames: [...]}}</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 20 |
| <a href="#">ObjectInfo</a>           | One entry of <code>listObjects</code> result.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 20 |
| <a href="#">ObjectState</a>          | Per-object snapshot returned by <code>getObjectsBatchState</code> . <code>properties</code> carries successful reads (one entry per property name resolved against the object's class hierarchy); <code>errors</code> is empty when nothing failed for this object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 21 |
| <a href="#">PropertyError</a>        | Per-property read error. Used when the caller-supplied <code>propertyNames</code> includes a name that doesn't exist on a given object's class, or when the accessor itself throws. Carried in a separate list from successful reads so a property literally named <code>error</code> can never collide with the failure envelope.                                                                                                                                                                                                                                                                                                                                                                                                                    | 21 |
| <a href="#">PropertyValuePair</a>    | Per-property changed value. The value type is decided by the addressed kernel object, not by JSON-RPC, so STL can't statically describe it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 21 |
| <a href="#">RegisteredBundleInfo</a> | Summary record returned by <code>listRegisteredBundles</code> . Excludes the file bytes; just enough to answer "what is currently being served, and where".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 21 |
| <a href="#">SessionInfo</a>          | One discovered session and the buses currently detected inside it. Names are opaque strings from the kernel's session-discovery surface.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 21 |
| <a href="#">StaticBundle</a>         | One bundle of in-memory files served on a configurable URL prefix. The same bundle's files are served at <code>&lt;urlPrefix&gt;/&lt;file.path&gt;</code> . <code>indexFileName</code> is the SPA fallback served whenever a request under <code>urlPrefix</code> does not match any file (lets the client-side router handle paths the server does not know about).                                                                                                                                                                                                                                                                                                                                                                                  | 22 |
| <a href="#">StaticFile</a>           | One file inside a <code>StaticBundle</code> . Bytes are owned by the bundle; <code>path</code> is relative to the bundle's <code>urlPrefix</code> and uses forward slashes regardless of the producer's OS. <code>contentType</code> is the MIME type the server will return verbatim; the bundle producer decides it (no extension-based inference happens server-side).                                                                                                                                                                                                                                                                                                                                                                             | 22 |
| <a href="#">SubscribeBlock</a>       | Pre-subscription block bundled into <code>createInterest</code> . Wires the listed property / event subscriptions on every matched object -- including ones that arrive later. Wire shape mirrors the typed STL representation exactly: each <code>MaybeMemberSelector</code> is either omitted (no auto-subscribe), <code>null</code> , or one of the typed variant arms ( <code>{type: "WildcardSelection", value: {}}</code> or <code>{type: "NamedSelection", value: {memberNames: [...]}}</code> ); <code>maxRateHz</code> is an optional positive number. Strict alignment with the typed surface lets the dispatcher's uniform meta-dispatch path consume the wire directly via <code>fromJson</code> , without a per-method translation step. | 22 |
| <a href="#">TlsConfig</a>            | TLS materials. The schema is reserved so configs stay forward-compatible, but the component fails to start if <code>tls</code> is set; full TLS support is not yet wired through the uWebSockets SSL backend.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 22 |
| <a href="#">TypeLookupResult</a>     | Bundled return of <code>getType</code> : always the <code>CustomTypeSpec</code> , plus the JSON-Schema fragment when the caller opted in. <code>schema</code> is empty when not requested.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 22 |
| <a href="#">WildcardSelection</a>    | Wildcard arm of <code>MemberSelector</code> : matches every member of the relevant kind. Wire shape: <code>{type: "WildcardSelection", value: {}}</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 23 |

## AddedObjectEntry STRUCTURE

One object newly entering an interest's match set.

| Name               | Type                              | Description                                                       |
|--------------------|-----------------------------------|-------------------------------------------------------------------|
| objectName         | <a href="#">string</a>            | name of the newly-matched object                                  |
| qualifiedClassName | <a href="#">string</a>            | package-qualified class name of the object                        |
| currentValues      | <a href="#">PropertyValueList</a> | initial values of pre-subscribed properties (empty if none wired) |

Named by [AddedObjectList](#).

## Configuration STRUCTURE

JSON-RPC over WebSocket component configuration. Only `address` and `port` are required; every other field falls back to a documented default.

| Name             | Type                             | Description                                                 |
|------------------|----------------------------------|-------------------------------------------------------------|
| address          | <a href="#">string</a>           | The IP address or hostname the server will bind to          |
| port             | <a href="#">u16</a>              | The port number the server will listen on                   |
| updateFreqHz     | <a href="#">f64</a>              | Run-loop frequency in Hz; defaults to 60                    |
| connectionLimits | <a href="#">ConnectionLimits</a> | Per-connection uWS buffer / timeout knobs                   |
| tls              | <a href="#">TlsConfig</a>        | schema placeholder; setting this fails component start      |
| controlBusName   | <a href="#">string</a>           | Control-plane bus on `local`; defaults to "jsonrpc_control" |

## ConnectionLimits STRUCTURE

Per-connection buffer / timeout knobs consulted on the uWS thread for the lifetime of the server. Every field is optional; unset fields fall back to the defaults resolved inside ws\_server.cpp at the use site.

| Name                  | Type                | Description                                                        |
|-----------------------|---------------------|--------------------------------------------------------------------|
| idleTimeoutSeconds    | <a href="#">u16</a> | uWS idle close timeout; defaults to 120                            |
| maxPayloadBytes       | <a href="#">u32</a> | Max inbound frame size; defaults to 16 KiB                         |
| maxBackpressureBytes  | <a href="#">u32</a> | uWS hard outbound buffer ceiling; defaults to 64 KiB               |
| highBackpressureBytes | <a href="#">u32</a> | Soft trigger for the BackpressureUpdate signal; defaults to 32 KiB |

Named by [Configuration](#).

## Identity STRUCTURE

Identity of the connected client. Set once on connect by the dispatcher from the WebSocket upgrade authenticator. Read-only externally. Other components / admin tools can `SELECT * FROM local.jsonrpc` to enumerate currently-connected clients.

| Name    | Type                   | Description                                                      |
|---------|------------------------|------------------------------------------------------------------|
| subject | <a href="#">string</a> | authenticated subject identifier (e.g. user / service principal) |

Named by [JsonRpcServer](#).

## NamedSelection STRUCTURE

Named arm of MemberSelector: matches only the listed members. Wire shape: `{type: "NamedSelection", value: {memberNames: [...]}}`.

| Name        | Type                                  | Description           |
|-------------|---------------------------------------|-----------------------|
| memberNames | <a href="#">sen.kernel.StringList</a> | member names to match |

Named by [MemberSelector](#).

## ObjectInfo STRUCTURE

One entry of listObjects' result.

| Name               | Type                   | Description                                 |
|--------------------|------------------------|---------------------------------------------|
| objectName         | <a href="#">string</a> | object name within the interest's match set |
| qualifiedClassName | <a href="#">string</a> | package-qualified class name of the object  |

Named by [ObjectInfos](#).

## ObjectState STRUCTURE

Per-object snapshot returned by ``getObjectsBatchState``. ``properties`` carries successful reads (one entry per property name resolved against the object's class hierarchy); ``errors`` is empty when nothing failed for this object.

| Name               | Type                              | Description                                                          |
|--------------------|-----------------------------------|----------------------------------------------------------------------|
| objectName         | <a href="#">string</a>            | object name within the interest's match set                          |
| qualifiedClassName | <a href="#">string</a>            | package-qualified class name at read time                            |
| properties         | <a href="#">PropertyValueList</a> | resolved property values; empty when caller's filter matched nothing |
| errors             | <a href="#">PropertyErrorList</a> | per-property failures; empty when none occurred                      |

Named by [ObjectStateList](#).

## PropertyError STRUCTURE

Per-property read error. Used when the caller-supplied ``propertyNames`` includes a name that doesn't exist on a given object's class, or when the accessor itself throws. Carried in a separate list from successful reads so a property literally named ``error`` can never collide with the failure envelope.

| Name         | Type                   | Description                                                                     |
|--------------|------------------------|---------------------------------------------------------------------------------|
| propertyName | <a href="#">string</a> | name from the caller's <code>`propertyNames`</code> list that could not be read |
| error        | <a href="#">string</a> | human-readable reason (e.g. "unknown property")                                 |

Named by [PropertyErrorList](#).

## PropertyValuePair STRUCTURE

Per-property changed value. The value type is decided by the addressed kernel object, not by JSON-RPC, so STL can't statically describe it.

| Name         | Type                   | Description           |
|--------------|------------------------|-----------------------|
| propertyName | <a href="#">string</a> | property member name  |
| value        | <a href="#">string</a> | JSON-encoded SenValue |

Named by [PropertyValueList](#).

## RegisteredBundleInfo STRUCTURE

Summary record returned by ``listRegisteredBundles``. Excludes the file bytes; just enough to answer "what is currently being served, and where".

| Name      | Type                   | Description                                                        |
|-----------|------------------------|--------------------------------------------------------------------|
| bundleId  | <a href="#">u64</a>    | opaque id assigned by the server, stable for the bundle's lifetime |
| urlPrefix | <a href="#">string</a> | bundle's URL prefix                                                |
| fileCount | <a href="#">u64</a>    | number of files in the bundle                                      |

Named by [RegisteredBundleInfoList](#).

## SessionInfo STRUCTURE

One discovered session and the buses currently detected inside it. Names are opaque strings from the kernel's session-discovery surface.

| Name  | Type                                  | Description                                  |
|-------|---------------------------------------|----------------------------------------------|
| name  | <a href="#">string</a>                | session name                                 |
| buses | <a href="#">sen.kernel.StringList</a> | bus names currently detected in this session |

Named by [SessionInfoList](#).

## StaticBundle STRUCTURE

One bundle of in-memory files served on a configurable URL prefix. The same bundle's files are served at `<urlPrefix>/<file.path>`. `indexFileName` is the SPA fallback served whenever a request under `urlPrefix` does not match any file (lets the client-side router handle paths the server does not know about).

| Name          | Type                           | Description                                                                                         |
|---------------|--------------------------------|-----------------------------------------------------------------------------------------------------|
| urlPrefix     | <a href="#">string</a>         | URL prefix, e.g. <code>"/explorer"</code> ; must start with <code>"/</code>                         |
| indexFileName | <a href="#">string</a>         | SPA fallback, e.g. <code>"index.html"</code> ; must match a <code>path</code> in <code>files</code> |
| files         | <a href="#">StaticFileList</a> | file set                                                                                            |

Named by [StaticFileServer](#).

## StaticFile STRUCTURE

One file inside a StaticBundle. Bytes are owned by the bundle; `path` is relative to the bundle's `urlPrefix` and uses forward slashes regardless of the producer's OS. `contentType` is the MIME type the server will return verbatim; the bundle producer decides it (no extension-based inference happens server-side).

| Name        | Type                              | Description                                                |
|-------------|-----------------------------------|------------------------------------------------------------|
| path        | <a href="#">string</a>            | path within the bundle, e.g. <code>"assets/main.js"</code> |
| contentType | <a href="#">string</a>            | MIME type, e.g. <code>"text/html"</code>                   |
| contents    | <a href="#">sen.kernel.Buffer</a> | raw file bytes                                             |

Named by [StaticFileList](#).

## SubscribeBlock STRUCTURE

Pre-subscription block bundled into `createInterest`. Wires the listed property / event subscriptions on every matched object -- including ones that arrive later. Wire shape mirrors the typed STL representation exactly: each `MaybeMemberSelector` is either omitted (no auto-subscribe), `null`, or one of the typed variant arms (`{type: "WildcardSelection", value: {}}` or `{type: "NamedSelection", value: {memberNames: [...]}}`); `maxRateHz` is an optional positive number. Strict alignment with the typed surface lets the dispatcher's uniform meta-dispatch path consume the wire directly via `fromJson`, without a per-method translation step.

| Name       | Type                           | Description                                                 |
|------------|--------------------------------|-------------------------------------------------------------|
| properties | <a href="#">MemberSelector</a> | which properties to auto-subscribe to (or none)             |
| events     | <a href="#">MemberSelector</a> | which events to auto-subscribe to (or none)                 |
| maxRateHz  | <a href="#">f64</a>            | optional throttle cap shared across the wired subscriptions |

Named by [JsonRpcServer](#).

## TlsConfig STRUCTURE

TLS materials. The schema is reserved so configs stay forward-compatible, but the component fails to start if `tls` is set; full TLS support is not yet wired through the `uWebSockets` SSL backend.

| Name     | Type                   | Description                    |
|----------|------------------------|--------------------------------|
| certPath | <a href="#">string</a> | PEM-encoded server certificate |
| keyPath  | <a href="#">string</a> | PEM-encoded server private key |

Named by [Configuration](#).

## TypeLookupResult STRUCTURE

Bundled return of `getType`: always the `CustomTypeSpec`, plus the JSON-Schema fragment when the caller opted in. `schema` is empty when not requested.

| Name | Type                                      | Description |
|------|-------------------------------------------|-------------|
| spec | <a href="#">sen.kernel.CustomTypeSpec</a> |             |

| Name   | Type                   | Description                                            |
|--------|------------------------|--------------------------------------------------------|
| schema | <a href="#">string</a> | JSON-encoded schema fragment; "" when withSchema=false |

Named by [JsonRpcServer](#).

## WildcardSelection STRUCTURE

Wildcard arm of MemberSelector: matches every member of the relevant kind. Wire shape: `{type: "WildcardSelection", value: {}}`.

Named by [MemberSelector](#).

### Variant records

[MemberSelector](#) Selector for properties or events in a SubscribeBlock. Absent = no auto-subscribe; WildcardSelection = "\*"; NamedSelection = explicit list. 23

## MemberSelector VARIANT

Selector for properties or events in a SubscribeBlock. Absent = no auto-subscribe; WildcardSelection = "\*"; NamedSelection = explicit list.

| Type                              | Description |
|-----------------------------------|-------------|
| <a href="#">WildcardSelection</a> |             |
| <a href="#">NamedSelection</a>    |             |

Named by [SubscribeBlock](#).

### Arrays

[AddedObjectList](#) 23  
[ObjectInfos](#) 23  
[ObjectStateList](#) 23  
[PropertyErrorList](#) 23  
[PropertyValueList](#) 24  
[RegisteredBundleInfoList](#) 24  
[SessionInfoList](#) 24  
[StaticFileList](#) 24

## AddedObjectList SEQUENCE

```
sequence<AddedObjectEntry> AddedObjectList;
```

Named by [JsonRpcServer](#).

## ObjectInfos SEQUENCE

```
sequence<ObjectInfo> ObjectInfos;
```

Named by [JsonRpcServer](#).

## ObjectStateList SEQUENCE

```
sequence<ObjectState> ObjectStateList;
```

Named by [JsonRpcServer](#).

## PropertyErrorList SEQUENCE

```
sequence<PropertyError> PropertyErrorList;
```

Named by [ObjectState](#).

## PropertyValueList SEQUENCE

```
sequence<PropertyValuePair> PropertyValueList;
```

Named by [AddedObjectEntry](#), [JsonRpcServer](#), [ObjectState](#).

## RegisteredBundleInfoList SEQUENCE

```
sequence<RegisteredBundleInfo> RegisteredBundleInfoList;
```

Named by [StaticFileServer](#).

## SessionInfoList SEQUENCE

```
sequence<SessionInfo> SessionInfoList;
```

Named by [JsonRpcServer](#).

## StaticFileList SEQUENCE

```
sequence<StaticFile> StaticFileList;
```

Named by [StaticBundle](#).

### Optionals

|                                                                                                                                                                                  |    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">MaybeConnectionLimits</a>                                                                                                                                            | 24 |
| <a href="#">MaybeControlBusName</a>                                                                                                                                              | 24 |
| <a href="#">MaybeFlag</a>                                                                                                                                                        | 24 |
| <a href="#">MaybeHighBackpressureBytes</a>                                                                                                                                       | 24 |
| <a href="#">MaybeIdleTimeoutSeconds</a>                                                                                                                                          | 25 |
| <a href="#">MaybeMaxBackpressureBytes</a>                                                                                                                                        | 25 |
| <a href="#">MaybeMaxPayloadBytes</a>                                                                                                                                             | 25 |
| <a href="#">MaybeMemberSelector</a>                                                                                                                                              | 25 |
| <a href="#">MaybeRateHz</a>                                                                                                                                                      | 25 |
| <a href="#">MaybeStringList</a>                                                                                                                                                  | 25 |
| Optional list of object names. Absent on the wire (or `null`) selects every object currently in the addressed interest's match set; present limits the read to the listed names. |    |
| <a href="#">MaybeSubscribeBlock</a>                                                                                                                                              | 25 |
| <a href="#">MaybeTlsConfig</a>                                                                                                                                                   | 25 |
| <a href="#">MaybeUpdateFreqHz</a>                                                                                                                                                | 25 |

## MaybeConnectionLimits OPTIONAL

```
optional<ConnectionLimits> MaybeConnectionLimits;
```

## MaybeControlBusName OPTIONAL

```
optional<string> MaybeControlBusName;
```

## MaybeFlag OPTIONAL

```
optional<bool> MaybeFlag;
```

## MaybeHighBackpressureBytes OPTIONAL

```
optional<u32> MaybeHighBackpressureBytes;
```

---

**MaybeIdleTimeoutSeconds** OPTIONAL

```
optional<u16> MaybeIdleTimeoutSeconds;
```

---

**MaybeMaxBackpressureBytes** OPTIONAL

```
optional<u32> MaybeMaxBackpressureBytes;
```

---

**MaybeMaxPayloadBytes** OPTIONAL

```
optional<u32> MaybeMaxPayloadBytes;
```

---

**MaybeMemberSelector** OPTIONAL

```
optional<MemberSelector> MaybeMemberSelector;
```

---

**MaybeRateHz** OPTIONAL

```
optional<f64> MaybeRateHz;
```

---

**MaybeStringList** OPTIONAL

Optional list of object names. Absent on the wire (or `null`) selects every object currently in the addressed interest's match set; present limits the read to the listed names.

```
optional<sen.kernel.StringList> MaybeStringList;
```

---

**MaybeSubscribeBlock** OPTIONAL

```
optional<SubscribeBlock> MaybeSubscribeBlock;
```

---

**MaybeTlsConfig** OPTIONAL

```
optional<TlsConfig> MaybeTlsConfig;
```

---

**MaybeUpdateFreqHz** OPTIONAL

```
optional<f64> MaybeUpdateFreqHz;
```

---

# logmaster

3 types.

## Object classes

[LogMaster](#) allows controlling the loggers  
[Logger](#) represents a logger

26

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## LogMaster CLASS

allows controlling the loggers

| Name      | Type                   | Flags   | Description |
|-----------|------------------------|---------|-------------|
| targetBus | <a href="#">string</a> | RO S BE |             |

| Method        | Signature                                         | Description                           |
|---------------|---------------------------------------------------|---------------------------------------|
| muteAll       | ()                                                | mute all the loggers                  |
| unmuteAll     | ()                                                | un-mutes all the muted loggers        |
| toggleMuteAll | ()                                                | toggles the muting of all loggers     |
| setLevel      | (level: <a href="#">sen.kernel.log.LogLevel</a> ) | sets all the loggers to a given level |

## Logger CLASS

represents a logger

| Name    | Type                                    | Flags   | Description                          |
|---------|-----------------------------------------|---------|--------------------------------------|
| level   | <a href="#">sen.kernel.log.LogLevel</a> | RO D BE | the current log level of this logger |
| pattern | <a href="#">string</a>                  | RO D C  | the pattern used by this logger      |
| muted   | <a href="#">bool</a>                    | RO D BE | true if muted                        |

| Method     | Signature                                         | Description                      |
|------------|---------------------------------------------------|----------------------------------|
| mute       | ()                                                | mute this logger                 |
| unmute     | ()                                                | unmute this logger               |
| setLevel   | (level: <a href="#">sen.kernel.log.LogLevel</a> ) | unmute this logger               |
| setPattern | (pattern: <a href="#">string</a> )                | sets the pattern for this logger |

## Fixed records

[Config](#) component configuration

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## Config STRUCTURE

component configuration

| Name      | Type                         | Description                  |
|-----------|------------------------------|------------------------------|
| targetBus | <a href="#">string</a>       | defaults to "local.log"      |
| period    | <a href="#">sen.Duration</a> | defaults to 500 milliseconds |

# py

3 types.

## Object classes

[PythonInterpreter](#) represents the embedded Python interpreter 27

## PythonInterpreter CLASS

represents the embedded Python interpreter

| Method   | Signature                                                | Description                   |
|----------|----------------------------------------------------------|-------------------------------|
| eval     | (expr: <a href="#">string</a> ) → <a href="#">string</a> | evaluates a Python expression |
| exec     | (code: <a href="#">string</a> )                          | executes a Python code block  |
| execFile | (file: <a href="#">string</a> )                          | executes a Python file        |

## Fixed records

[Configuration](#) Overall component configuration 27

## Configuration STRUCTURE

Overall component configuration

| Name    | Type                       | Description                                                     |
|---------|----------------------------|-----------------------------------------------------------------|
| freqHz  | <a href="#">f32</a>        | update frequency in Hertz                                       |
| module  | <a href="#">string</a>     | python module to use (name of a python script, without the .py) |
| imports | <a href="#">StringList</a> | sen packages to import                                          |
| bus     | <a href="#">string</a>     | bus where to publish the interpreter object (none by default)   |

## Arrays

[StringList](#) 27

## StringList SEQUENCE

```
sequence<string> StringList;
```

Named by [Configuration](#).

# recorder

6 types.

## Object classes

[Recorder](#) represents an recording session

28

## Recorder CLASS

represents an recording session

| Name     | Type                              | Flags   | Description |
|----------|-----------------------------------|---------|-------------|
| settings | <a href="#">RecordingSettings</a> | RO S BE |             |
| state    | <a href="#">RecorderState</a>     | RO D C  |             |

| Method     | Signature                            | Description                   |
|------------|--------------------------------------|-------------------------------|
| start      | ()                                   | starts (or resumes) recording |
| stop       | ()                                   | stops recording               |
| fetchStats | () → <a href="#">sen.db.OutStats</a> | sample the output statistics  |

## Fixed records

[Configuration](#) Overall component configuration

28

[RecordingSettings](#) Configuration of a predefined recording

28

## Configuration STRUCTURE

Overall component configuration

| Name           | Type                                  | Description                                      |
|----------------|---------------------------------------|--------------------------------------------------|
| samplingPeriod | <a href="#">sen.Duration</a>          | sampling period for the recordings               |
| recordings     | <a href="#">RecordingSettingsList</a> | predefined recordings                            |
| bus            | <a href="#">string</a>                | bus to publish the object (local.rec by default) |

## RecordingSettings STRUCTURE

Configuration of a predefined recording

| Name           | Type                          | Description                                     |
|----------------|-------------------------------|-------------------------------------------------|
| name           | <a href="#">string</a>        | name of the recording (cannot be empty)         |
| folder         | <a href="#">string</a>        | storage folder                                  |
| selections     | <a href="#">SelectionList</a> | which objects shall be recorded                 |
| keyframePeriod | <a href="#">sen.Duration</a>  | period between keyframes (0 means no keyframes) |
| indexKeyframes | <a href="#">bool</a>          | should we create indexes for keyframes          |
| indexObjects   | <a href="#">bool</a>          | should we create indexes for objects            |
| autoStart      | <a href="#">bool</a>          | should we start recording automatically         |

Named by [Recorder](#), [RecordingSettingsList](#).

## Enumerations

[RecorderState](#) the state of a recording session

28

## RecorderState ENUMERATION

the state of a recording session

Held as u8.

|           |   |       |   |
|-----------|---|-------|---|
| stopped   | 0 | error | 2 |
| recording | 1 |       |   |

Named by [Recorder](#).

Arrays

|                                       |                               |    |
|---------------------------------------|-------------------------------|----|
| <a href="#">RecordingSettingsList</a> | List of predefined recordings | 29 |
| <a href="#">SelectionList</a>         | List of selection queries     | 29 |

---

**RecordingSettingsList** SEQUENCE

List of predefined recordings

```
sequence<RecordingSettings> RecordingSettingsList;
```

Named by [Configuration](#).

---

**SelectionList** SEQUENCE

List of selection queries

```
sequence<string> SelectionList;
```

Named by [RecordingSettings](#).

# replayer

4 types.

## Object classes

|                          |                                                               |    |
|--------------------------|---------------------------------------------------------------|----|
| <a href="#">Replay</a>   | controls the replay of a given archive                        | 30 |
| <a href="#">Replayer</a> | represents the replayer component and manages Replay objects. | 30 |

## Replay CLASS

controls the replay of a given archive

| Name         | Type                           | Flags   | Description                                  |
|--------------|--------------------------------|---------|----------------------------------------------|
| archiveInfo  | <a href="#">sen.db.Summary</a> | RO S BE | information about the archive being replayed |
| archivePath  | <a href="#">string</a>         | RO S BE | file path to the archive                     |
| status       | <a href="#">ReplayStatus</a>   | RO D C  | playback status                              |
| playbackTime | <a href="#">sen.Timestamp</a>  | RO D BE | playback time                                |

| Method  | Signature                              | Description                                                          |
|---------|----------------------------------------|----------------------------------------------------------------------|
| play    | ()                                     | starts the playback from the current position                        |
| pause   | ()                                     | hold the playback position, if playing                               |
| stop    | ()                                     | goes to the start of the playback, erasing all objects               |
| seek    | (time: <a href="#">sen.Timestamp</a> ) | restores the playback to the keyframe that's closest to a given time |
| advance | (time: <a href="#">sen.Duration</a> )  | if paused or stopped, advances the playback a given time delta       |

## Replayer CLASS

represents the replayer component and manages Replay objects.

| Name     | Type                 | Flags   | Description                                             |
|----------|----------------------|---------|---------------------------------------------------------|
| autoPlay | <a href="#">bool</a> | RO S BE | if true, automatically calls play when opening a replay |

| Method   | Signature                                                      | Description                                                          |
|----------|----------------------------------------------------------------|----------------------------------------------------------------------|
| open     | (name: <a href="#">string</a> , path: <a href="#">string</a> ) | opens an archive in a given path, and creates a named Replay object. |
| close    | (name: <a href="#">string</a> )                                | closes the named Replay object, if any.                              |
| closeAll | ()                                                             | closes all previously opened replays                                 |

## Fixed records

|                               |                                 |    |
|-------------------------------|---------------------------------|----|
| <a href="#">Configuration</a> | Overall component configuration | 30 |
|-------------------------------|---------------------------------|----|

## Configuration STRUCTURE

Overall component configuration

| Name           | Type                         | Description                                                             |
|----------------|------------------------------|-------------------------------------------------------------------------|
| bus            | <a href="#">string</a>       | bus to publish the object (local.replay by default)                     |
| samplingPeriod | <a href="#">sen.Duration</a> | max sampling period for flushing the replayed data                      |
| autoOpen       | <a href="#">string</a>       | optional archive path to auto open on start-up (empty by default)       |
| autoPlay       | <a href="#">bool</a>         | if autoOpen is not empty, try to play it on start-up (false by default) |

Enumerations

ReplayStatus

 the state in which a Replay can be 31

ReplayStatus **ENUMERATION**

the state in which a Replay can be

Held as u8.

|         |   |
|---------|---|
| stopped | 0 |
| paused  | 1 |
| playing | 2 |

Named by [Replay](#).

# rest

37 types.

## Fixed records

|                                     |    |
|-------------------------------------|----|
| <a href="#">Argument</a>            | 32 |
| <a href="#">Bus</a>                 | 32 |
| <a href="#">BusSummary</a>          | 32 |
| <a href="#">ClientAuth</a>          | 32 |
| <a href="#">Configuration</a>       | 33 |
| <a href="#">Error</a>               | 33 |
| <a href="#">Interest</a>            | 33 |
| <a href="#">InterestSummary</a>     | 33 |
| <a href="#">Link</a>                | 33 |
| <a href="#">Notification</a>        | 33 |
| <a href="#">Object</a>              | 33 |
| <a href="#">ObjectEvent</a>         | 34 |
| <a href="#">ObjectMethod</a>        | 34 |
| <a href="#">ObjectSummary</a>       | 34 |
| <a href="#">Session</a>             | 34 |
| <a href="#">SessionSummary</a>      | 34 |
| <a href="#">SubscriptionOptions</a> | 34 |
| <a href="#">Subscriptions</a>       | 34 |
| <a href="#">Success</a>             | 35 |
| <a href="#">Version</a>             | 35 |

## Argument STRUCTURE

| Name        | Type                   | Description |
|-------------|------------------------|-------------|
| name        | <a href="#">string</a> |             |
| description | <a href="#">string</a> |             |
| type        | <a href="#">string</a> |             |

Named by [Arguments](#).

## Bus STRUCTURE

| Name    | Type                           | Description |
|---------|--------------------------------|-------------|
| id      | <a href="#">string</a>         |             |
| status  | <a href="#">Status</a>         |             |
| objects | <a href="#">ObjectsSummary</a> |             |

## BusSummary STRUCTURE

| Name   | Type                   | Description |
|--------|------------------------|-------------|
| id     | <a href="#">string</a> |             |
| status | <a href="#">Status</a> |             |

Named by [Buses](#).

## ClientAuth STRUCTURE

| Name | Type                   | Description |
|------|------------------------|-------------|
| id   | <a href="#">string</a> |             |

## Configuration STRUCTURE

| Name    | Type                   | Description                                        |
|---------|------------------------|----------------------------------------------------|
| address | <a href="#">string</a> | The IP address or hostname the server will bind to |
| port    | <a href="#">u16</a>    | The port number the server will listen on          |
| freqHz  | <a href="#">f32</a>    | Update frequency in Hertz                          |

## Error STRUCTURE

| Name  | Type                   | Description |
|-------|------------------------|-------------|
| error | <a href="#">string</a> |             |

## Interest STRUCTURE

| Name  | Type                   | Description |
|-------|------------------------|-------------|
| query | <a href="#">string</a> |             |
| name  | <a href="#">string</a> |             |

## InterestSummary STRUCTURE

| Name    | Type                   | Description |
|---------|------------------------|-------------|
| name    | <a href="#">string</a> |             |
| session | <a href="#">string</a> |             |
| bus     | <a href="#">string</a> |             |

Named by [InterestsSummary](#).

## Link STRUCTURE

| Name   | Type                       | Description |
|--------|----------------------------|-------------|
| rel    | <a href="#">RelType</a>    |             |
| href   | <a href="#">string</a>     |             |
| method | <a href="#">HttpMethod</a> |             |

Named by [Links](#), [ObjectSummary](#).

## Notification STRUCTURE

| Name     | Type                             | Description |
|----------|----------------------------------|-------------|
| type     | <a href="#">NotificationType</a> |             |
| interest | <a href="#">string</a>           |             |
| time     | <a href="#">sen.TimeStamp</a>    |             |
| data     | <a href="#">string</a>           |             |

## Object STRUCTURE

| Name      | Type                   | Description |
|-----------|------------------------|-------------|
| objectId  | <a href="#">u32</a>    |             |
| name      | <a href="#">string</a> |             |
| className | <a href="#">string</a> |             |
| localName | <a href="#">string</a> |             |

| Name        | Type                   | Description |
|-------------|------------------------|-------------|
| description | <a href="#">string</a> |             |
| links       | <a href="#">Links</a>  |             |

## ObjectEvent STRUCTURE

| Name        | Type                   | Description |
|-------------|------------------------|-------------|
| name        | <a href="#">string</a> |             |
| description | <a href="#">string</a> |             |

## ObjectMethod STRUCTURE

| Name        | Type                      | Description |
|-------------|---------------------------|-------------|
| name        | <a href="#">string</a>    |             |
| description | <a href="#">string</a>    |             |
| retType     | <a href="#">string</a>    |             |
| args        | <a href="#">Arguments</a> |             |

## ObjectSummary STRUCTURE

| Name      | Type                   | Description |
|-----------|------------------------|-------------|
| objectId  | <a href="#">u32</a>    |             |
| name      | <a href="#">string</a> |             |
| className | <a href="#">string</a> |             |
| localName | <a href="#">string</a> |             |
| link      | <a href="#">Link</a>   |             |

Named by [ObjectsSummary](#).

## Session STRUCTURE

| Name | Type                   | Description |
|------|------------------------|-------------|
| name | <a href="#">string</a> |             |

## SessionSummary STRUCTURE

| Name  | Type                     | Description |
|-------|--------------------------|-------------|
| name  | <a href="#">string</a>   |             |
| buses | <a href="#">BusNames</a> |             |

## SubscriptionOptions STRUCTURE

| Name          | Type                         | Description      |
|---------------|------------------------------|------------------|
| maxUpdateTime | <a href="#">sen.Duration</a> | Max. update time |

## Subscriptions STRUCTURE

| Name       | Type                       | Description |
|------------|----------------------------|-------------|
| properties | <a href="#">Properties</a> |             |
| events     | <a href="#">Events</a>     |             |

Success STRUCTURE

| Name | Type   | Description |
|------|--------|-------------|
| msg  | string |             |

Version STRUCTURE

| Name    | Type   | Description |
|---------|--------|-------------|
| version | string |             |

Enumerations

|                  |    |
|------------------|----|
| HttpMethod       | 35 |
| InvokeStatus     | 35 |
| NotificationType | 35 |
| RelType          | 35 |
| Status           | 36 |

HttpMethod ENUMERATION

Held as u8.

|            |   |
|------------|---|
| httpGet    | 0 |
| httpPost   | 1 |
| httpPut    | 2 |
| httpDelete | 3 |
| httpOption | 4 |

Named by [Link](#).

InvokeStatus ENUMERATION

Held as u8.

|          |   |
|----------|---|
| pending  | 0 |
| finished | 1 |
| failed   | 2 |

NotificationType ENUMERATION

Held as u8.

|               |   |
|---------------|---|
| evt           | 0 |
| invoke        | 1 |
| property      | 2 |
| objectAdded   | 3 |
| objectRemoved | 4 |
| count         | 5 |

Named by [Notification](#).

RelType ENUMERATION

Held as u8.

|                     |   |
|---------------------|---|
| method              | 0 |
| setter              | 1 |
| getter              | 2 |
| evt                 | 3 |
| property            | 4 |
| propertySubscribe   | 5 |
| propertyUnsubscribe | 6 |
| def                 | 7 |

Named by [Link](#).

Status

ENUMERATION

Held as u8.

|       |   |
|-------|---|
| open  | 0 |
| close | 1 |

Named by [Bus](#), [BusSummary](#).

Arrays

|                                  |    |
|----------------------------------|----|
| <a href="#">Arguments</a>        | 36 |
| <a href="#">BusNames</a>         | 36 |
| <a href="#">Buses</a>            | 36 |
| <a href="#">Events</a>           | 36 |
| <a href="#">InterestsSummary</a> | 36 |
| <a href="#">Links</a>            | 36 |
| <a href="#">ObjectsSummary</a>   | 36 |
| <a href="#">Properties</a>       | 37 |
| <a href="#">Sessions</a>         | 37 |

Arguments

SEQUENCE

```
sequence<Argument> Arguments;
```

Named by [ObjectMethod](#).

BusNames

SEQUENCE

```
sequence<string> BusNames;
```

Named by [SessionSummary](#).

Buses

SEQUENCE

```
sequence<BusSummary> Buses;
```

Events

SEQUENCE

```
sequence<string> Events;
```

Named by [Subscriptions](#).

InterestsSummary

SEQUENCE

```
sequence<InterestSummary> InterestsSummary;
```

Links

SEQUENCE

```
sequence<Link> Links;
```

Named by [Object](#).

ObjectsSummary

SEQUENCE

```
sequence<ObjectSummary> ObjectsSummary;
```

Named by [Bus](#).

---

## Properties SEQUENCE

```
sequence<string> Properties;
```

Named by [Subscriptions](#).

---

## Sessions SEQUENCE

```
sequence<string> Sessions;
```

### Optionals

|                                 |    |
|---------------------------------|----|
| <a href="#">ThreadPoolSize</a>  | 37 |
| <a href="#">UpdateFrequency</a> | 37 |
| <a href="#">UpdateTime</a>      | 37 |

---

## ThreadPoolSize OPTIONAL

```
optional<u16> ThreadPoolSize;
```

---

## UpdateFrequency OPTIONAL

```
optional<f32> UpdateFrequency;
```

---

## UpdateTime OPTIONAL

```
optional<sen.Duration> UpdateTime;
```

# shell

27 types.

## Object classes

[Shell](#) Provides a text-based interface to interact with the a sen kernel

38

## Shell CLASS

Provides a text-based interface to interact with the a sen kernel

| Name   | Type                          | Flags   | Description           |
|--------|-------------------------------|---------|-----------------------|
| config | <a href="#">Configuration</a> | RO S BE | service configuration |

| Method   | Signature                                                              | Description                                               |
|----------|------------------------------------------------------------------------|-----------------------------------------------------------|
| clear    | ()                                                                     | clears the screen                                         |
| help     | ()                                                                     | basic usage help                                          |
| history  | ()                                                                     | lists previous commands                                   |
| ls       | ()                                                                     | lists all the existing objects and sources.               |
| open     | (source: <a href="#">string</a> )                                      | opens a source (bus or session) without any condition     |
| query    | (name: <a href="#">string</a> ,<br>condition: <a href="#">string</a> ) | opens a named source using a condition expression         |
| close    | (source: <a href="#">string</a> )                                      | closes a previously-opened source (session, bus or query) |
| info     | (term: <a href="#">string</a> )                                        | prints information about a type or instance               |
| shutdown | ()                                                                     | requests the kernel to be shut-down                       |
| src      | ()                                                                     | lists the current sources                                 |

## Fixed records

|                                           |                                       |    |
|-------------------------------------------|---------------------------------------|----|
| <a href="#">CPrint</a>                    |                                       | 38 |
| <a href="#">ClearCurrentLine</a>          |                                       | 39 |
| <a href="#">ClearRemainingCurrentLine</a> |                                       | 39 |
| <a href="#">ClearScreen</a>               |                                       | 39 |
| <a href="#">Configuration</a>             | Configuration params for the shell    | 39 |
| <a href="#">HideCursor</a>                |                                       | 39 |
| <a href="#">MoveCursorAllLeft</a>         |                                       | 39 |
| <a href="#">MoveCursorDown</a>            |                                       | 39 |
| <a href="#">MoveCursorLeft</a>            |                                       | 39 |
| <a href="#">MoveCursorRight</a>           |                                       | 39 |
| <a href="#">MoveCursorUp</a>              |                                       | 40 |
| <a href="#">NewLine</a>                   |                                       | 40 |
| <a href="#">Print</a>                     |                                       | 40 |
| <a href="#">Query</a>                     | Holds information about a named query | 40 |
| <a href="#">RestoreCursorPosition</a>     |                                       | 40 |
| <a href="#">SaveCursorPosition</a>        |                                       | 40 |
| <a href="#">SetBgColor</a>                |                                       | 40 |
| <a href="#">SetFgColor</a>                |                                       | 40 |
| <a href="#">SetWindowTitle</a>            |                                       | 40 |
| <a href="#">ShowCursor</a>                |                                       | 41 |

## CPrint STRUCTURE

| Name  | Type                   | Description |
|-------|------------------------|-------------|
| flags | <a href="#">u32</a>    |             |
| color | <a href="#">u32</a>    |             |
| text  | <a href="#">string</a> |             |

Named by [TerminalCmd](#).

## ClearCurrentLine STRUCTURE

Named by [TerminalCmd](#).

## ClearRemainingCurrentLine STRUCTURE

Named by [TerminalCmd](#).

## ClearScreen STRUCTURE

Named by [TerminalCmd](#).

## Configuration STRUCTURE

Configuration params for the shell

| Name          | Type                        | Description                                              |
|---------------|-----------------------------|----------------------------------------------------------|
| prompt        | <a href="#">string</a>      | defaults to 'sen> '                                      |
| timeStyle     | <a href="#">TimeStyle</a>   | how to display time                                      |
| bufferStyle   | <a href="#">BufferStyle</a> | how to display buffers                                   |
| serverEnabled | <a href="#">bool</a>        | enables remote shell access (false by default)           |
| serverPort    | <a href="#">u16</a>         | 0 means default                                          |
| noLogo        | <a href="#">bool</a>        | true if you don't want a welcoming logo                  |
| open          | <a href="#">SourcesList</a> | list of sources to open automatically                    |
| query         | <a href="#">QueryList</a>   | list of queries to create automatically                  |
| logBus        | <a href="#">string</a>      | where to find the logmaster, if any (default: local.log) |

Named by [Shell](#).

## HideCursor STRUCTURE

Named by [TerminalCmd](#).

## MoveCursorAllLeft STRUCTURE

Named by [TerminalCmd](#).

## MoveCursorDown STRUCTURE

| Name  | Type                | Description |
|-------|---------------------|-------------|
| cells | <a href="#">u32</a> |             |

Named by [TerminalCmd](#).

## MoveCursorLeft STRUCTURE

| Name  | Type                | Description |
|-------|---------------------|-------------|
| cells | <a href="#">u32</a> |             |

Named by [TerminalCmd](#).

## MoveCursorRight STRUCTURE

| Name  | Type                | Description |
|-------|---------------------|-------------|
| cells | <a href="#">u32</a> |             |

Named by [TerminalCmd](#).

## MoveCursorUp STRUCTURE

| Name  | Type | Description |
|-------|------|-------------|
| cells | u32  |             |

Named by [TerminalCmd](#).

## NewLine STRUCTURE

Named by [TerminalCmd](#).

## Print STRUCTURE

| Name | Type   | Description |
|------|--------|-------------|
| text | string |             |

Named by [TerminalCmd](#).

## Query STRUCTURE

Holds information about a named query

| Name      | Type   | Description                                           |
|-----------|--------|-------------------------------------------------------|
| name      | string | the name of the query (cannot contain spaces or dots) |
| selection | string | the query expression                                  |

Named by [QueryList](#).

## RestoreCursorPosition STRUCTURE

Named by [TerminalCmd](#).

## SaveCursorPosition STRUCTURE

Named by [TerminalCmd](#).

## SetBgColor STRUCTURE

| Name  | Type | Description |
|-------|------|-------------|
| color | u32  |             |

Named by [TerminalCmd](#).

## SetFgColor STRUCTURE

| Name  | Type | Description |
|-------|------|-------------|
| color | u32  |             |

Named by [TerminalCmd](#).

## SetWindowTitle STRUCTURE

| Name | Type   | Description |
|------|--------|-------------|
| text | string |             |

Named by [TerminalCmd](#).

## ShowCursor STRUCTURE

Named by [TerminalCmd](#).

### Enumerations

|                             |                                  |    |
|-----------------------------|----------------------------------|----|
| <a href="#">BufferStyle</a> | How buffers should be printed    | 41 |
| <a href="#">ListStyle</a>   | How lists should be printed      | 41 |
| <a href="#">TimeStyle</a>   | How timestamps should be printed | 41 |

## BufferStyle ENUMERATION

How buffers should be printed

Held as u8.

|         |   |
|---------|---|
| size    | 0 |
| hexdump | 1 |

Named by [Configuration](#).

## ListStyle ENUMERATION

How lists should be printed

Held as u8.

|            |   |
|------------|---|
| treeStyle  | 0 |
| plainStyle | 1 |

## TimeStyle ENUMERATION

How timestamps should be printed

Held as u8.

|                |   |
|----------------|---|
| timestampUTC   | 0 |
| timestampLocal | 1 |

Named by [Configuration](#).

### Variant records

|                             |    |
|-----------------------------|----|
| <a href="#">TerminalCmd</a> | 41 |
|-----------------------------|----|

## TerminalCmd VARIANT

| Type                                      | Description |
|-------------------------------------------|-------------|
| <a href="#">Print</a>                     |             |
| <a href="#">CPrint</a>                    |             |
| <a href="#">NewLine</a>                   |             |
| <a href="#">MoveCursorLeft</a>            |             |
| <a href="#">MoveCursorRight</a>           |             |
| <a href="#">MoveCursorUp</a>              |             |
| <a href="#">MoveCursorDown</a>            |             |
| <a href="#">MoveCursorAllLeft</a>         |             |
| <a href="#">SetFgColor</a>                |             |
| <a href="#">SetBgColor</a>                |             |
| <a href="#">ClearCurrentLine</a>          |             |
| <a href="#">ClearRemainingCurrentLine</a> |             |
| <a href="#">HideCursor</a>                |             |
| <a href="#">ShowCursor</a>                |             |
| <a href="#">SaveCursorPosition</a>        |             |

| Type                                  | Description |
|---------------------------------------|-------------|
| <a href="#">RestoreCursorPosition</a> |             |
| <a href="#">ClearScreen</a>           |             |
| <a href="#">SetWindowTitle</a>        |             |

Arrays

|                             |    |
|-----------------------------|----|
| <a href="#">QueryList</a>   | 42 |
| <a href="#">SourcesList</a> | 42 |

QueryList SEQUENCE

```
sequence<Query> QueryList;
```

Named by [Configuration](#).

SourcesList SEQUENCE

```
sequence<string> SourcesList;
```

Named by [Configuration](#).

# db

3 types.

## Fixed records

|                             |                       |    |
|-----------------------------|-----------------------|----|
| <a href="#">OutSettings</a> | Settings for outputs  | 43 |
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## OutSettings STRUCTURE

Settings for outputs

| Name           | Type                   | Description                            |
|----------------|------------------------|----------------------------------------|
| name           | <a href="#">string</a> | name of the archive (cannot be empty)  |
| folder         | <a href="#">string</a> | storage folder                         |
| indexKeyframes | <a href="#">bool</a>   | should we create indexes for keyframes |

## OutStats STRUCTURE

Statistics of writing

| Name         | Type                | Description |
|--------------|---------------------|-------------|
| maxQueueSize | <a href="#">u64</a> | in bytes    |

Named by [sen.components.recorder.Recorder](#).

## Summary STRUCTURE

Summary of an archive

| Name               | Type                          | Description                 |
|--------------------|-------------------------------|-----------------------------|
| firstTime          | <a href="#">sen.TimeStamp</a> | the time of the first event |
| lastTime           | <a href="#">sen.TimeStamp</a> | the time of the last event  |
| keyframeCount      | <a href="#">u32</a>           | number of written keyframes |
| objectCount        | <a href="#">u32</a>           | number of recorded objects  |
| typeCount          | <a href="#">u32</a>           | number of types written     |
| annotationCount    | <a href="#">u32</a>           | number of annotations       |
| indexedObjectCount | <a href="#">u32</a>           | number of indexed objects   |

Named by [sen.components.replayer.Replay](#).

# kernel

139 types.

## Object classes

|                                    |                                                                                                                                                                                                                 |    |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">KernelApi</a>          | Represents the kernel                                                                                                                                                                                           | 44 |
| <a href="#">VirtualClock</a>       | Allows inspecting and controlling the advance of virtual time and associated component iteration                                                                                                                | 44 |
| <a href="#">VirtualKernelClock</a> | A clock for all the components loaded by the kernel                                                                                                                                                             | 44 |
| <a href="#">VirtualMasterClock</a> | Controls all the VirtualKernelClock that it discovers in the bus where it's published. One step reaches the next component due to run: each clock reports how long until its own next one, and the soonest wins | 44 |

## KernelApi CLASS

Represents the kernel

| Name      | Type                      | Flags   | Description              |
|-----------|---------------------------|---------|--------------------------|
| buildInfo | <a href="#">BuildInfo</a> | RO S BE | kernel build information |

  

| Method    | Signature                         | Description                           |
|-----------|-----------------------------------|---------------------------------------|
| shutdown  | ()                                | requests the kernel to shut down      |
| getUnits  | () → <a href="#">UnitList</a>     | registered units                      |
| getTypes  | () → <a href="#">StringList</a>   | registered types, by name             |
| getConfig | () → <a href="#">KernelParams</a> | configuration of the kernel execution |

## VirtualClock CLASS

Allows inspecting and controlling the advance of virtual time and associated component iteration

| Name | Type                          | Flags   | Description              |
|------|-------------------------------|---------|--------------------------|
| time | <a href="#">sen.Timestamp</a> | RO D BE | the current virtual time |

## VirtualKernelClock CLASS

A clock for all the components loaded by the kernel

| Method         | Signature                                                             | Description                                                                            |
|----------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| processNoFlush | (delta: <a href="#">sen.Duration</a> ) → <a href="#">sen.Duration</a> | updates the time, drains the inputs, and cycles components - doesn't flush outputs     |
| flushOutputs   | () → <a href="#">sen.Duration</a>                                     | makes component outputs visible to other components and returns the next step duration |

## VirtualMasterClock CLASS

Controls all the VirtualKernelClock that it discovers in the bus where it's published. One step reaches the next component due to run: each clock reports how long until its own next one, and the soonest wins

| Method      | Signature                                                                | Description                                                                                                              |
|-------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| advanceTime | (duration: <a href="#">sen.Duration</a> ) → <a href="#">sen.Duration</a> | steps until the virtual time advanced reaches duration, rounding up to a whole step. Returns the wall time the call took |
| step        | () → <a href="#">sen.Duration</a>                                        | single step forward                                                                                                      |
| steps       | (count: <a href="#">u64</a> ) → <a href="#">sen.Duration</a>             | perform 'count' steps forward                                                                                            |

## Fixed records

|                                 |                                                |    |
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| <a href="#">AliasTypeSpec</a>   | data of an aliased type                        | 46 |
| <a href="#">AliasTypeSpecV4</a> |                                                | 46 |
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| <a href="#">ArgSpec</a>         | an argument of a callable (methods and events) | 46 |

|                                        |                                                                     |    |
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| <a href="#">ArgSpecV5</a>              |                                                                     | 46 |
| <a href="#">BuildInfo</a>              | Build-related information                                           | 47 |
| <a href="#">BuiltComponentParams</a>   | Parameters related to components build by the kernel                | 47 |
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| <a href="#">ClassTypeSpec</a>          | spec of a class                                                     | 47 |
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| <a href="#">ClassTypeSpecV5</a>        |                                                                     | 48 |
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| <a href="#">SequenceTypeSpecV4</a>     |                                                                     | 56 |
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## AliasTypeSpec STRUCTURE

data of an aliased type

| Name        | Type                   | Description                       |
|-------------|------------------------|-----------------------------------|
| aliasedType | <a href="#">string</a> | qualified type name being aliased |

Named by [CustomTypeData](#).

## AliasTypeSpecV4 STRUCTURE

| Name        | Type                   | Description                       |
|-------------|------------------------|-----------------------------------|
| aliasedType | <a href="#">string</a> | qualified type name being aliased |

Named by [CustomTypeDataV4](#).

## AliasTypeSpecV5 STRUCTURE

| Name        | Type                   | Description                       |
|-------------|------------------------|-----------------------------------|
| aliasedType | <a href="#">string</a> | qualified type name being aliased |
| hash        | <a href="#">u32</a>    | unique hash                       |

Named by [CustomTypeDataV5](#).

## ArgSpec STRUCTURE

an argument of a callable (methods and events)

| Name        | Type                   | Description                         |
|-------------|------------------------|-------------------------------------|
| name        | <a href="#">string</a> | argument name                       |
| description | <a href="#">string</a> | documentation                       |
| type        | <a href="#">string</a> | qualified type name of the argument |

Named by [ArgSpecList](#).

## ArgSpecV4 STRUCTURE

| Name        | Type                   | Description                         |
|-------------|------------------------|-------------------------------------|
| name        | <a href="#">string</a> | argument name                       |
| description | <a href="#">string</a> | documentation                       |
| type        | <a href="#">string</a> | qualified type name of the argument |

Named by [ArgSpecListV4](#).

## ArgSpecV5 STRUCTURE

| Name        | Type                   | Description                          |
|-------------|------------------------|--------------------------------------|
| name        | <a href="#">string</a> | argument name                        |
| nameHash    | <a href="#">u32</a>    | hash computed with the argument name |
| description | <a href="#">string</a> | documentation                        |
| type        | <a href="#">string</a> | qualified type name of the argument  |
| hash        | <a href="#">u32</a>    | unique hash                          |

Named by [ArgSpecListV5](#).

## BuildInfo STRUCTURE

Build-related information

| Name       | Type                      | Description                                                |
|------------|---------------------------|------------------------------------------------------------|
| maintainer | <a href="#">string</a>    | principal maintainer of the software                       |
| version    | <a href="#">string</a>    | version string (format-agnostic)                           |
| compiler   | <a href="#">string</a>    | vendor-specific compiler string                            |
| debugMode  | <a href="#">bool</a>      | compiled in debug mode or not                              |
| buildTime  | <a href="#">string</a>    | the commit's timestamp, ISO-8601; not the compiler's clock |
| wordSize   | <a href="#">WordSize</a>  | architecture                                               |
| gitRef     | <a href="#">string</a>    | git ref spec                                               |
| gitHash    | <a href="#">string</a>    | git hash                                                   |
| gitStatus  | <a href="#">GitStatus</a> | git status                                                 |

Named by [ComponentInfo](#), [KernelApi](#).

## BuiltComponentParams STRUCTURE

Parameters related to components build by the kernel

| Name    | Type                            | Description                        |
|---------|---------------------------------|------------------------------------|
| name    | <a href="#">string</a>          | the name of the component          |
| period  | <a href="#">sen.Duration</a>    | the execution period               |
| config  | <a href="#">ComponentConfig</a> | the configuration of the component |
| imports | <a href="#">StringList</a>      | the imported packages              |

Named by [BuiltComponentList](#).

## BusAddress STRUCTURE

Address to connect to a bus

| Name        | Type                   | Description |
|-------------|------------------------|-------------|
| sessionName | <a href="#">string</a> |             |
| busName     | <a href="#">string</a> |             |

## ClassTypeSpec STRUCTURE

spec of a class

| Name        | Type                             | Description |
|-------------|----------------------------------|-------------|
| properties  | <a href="#">PropertySpecList</a> |             |
| methods     | <a href="#">MethodSpecList</a>   |             |
| events      | <a href="#">EventSpecList</a>    |             |
| constructor | <a href="#">MethodSpec</a>       |             |
| parents     | <a href="#">StringList</a>       |             |
| isInterface | <a href="#">bool</a>             |             |

Named by [CustomTypeData](#).

## ClassTypeSpecV4 STRUCTURE

| Name       | Type                               | Description |
|------------|------------------------------------|-------------|
| properties | <a href="#">PropertySpecListV4</a> |             |
| methods    | <a href="#">MethodSpecListV4</a>   |             |
| events     | <a href="#">EventSpecListV4</a>    |             |

| Name        | Type                         | Description |
|-------------|------------------------------|-------------|
| isInterface | <a href="#">bool</a>         |             |
| id          | <a href="#">u32</a>          |             |
| constructor | <a href="#">MethodSpecV4</a> |             |
| parents     | <a href="#">StringList</a>   |             |

Named by [CustomTypeDataV4](#).

## ClassTypeSpecV5 STRUCTURE

| Name        | Type                               | Description |
|-------------|------------------------------------|-------------|
| properties  | <a href="#">PropertySpecListV5</a> |             |
| methods     | <a href="#">MethodSpecListV5</a>   |             |
| events      | <a href="#">EventSpecListV5</a>    |             |
| isInterface | <a href="#">bool</a>               |             |
| id          | <a href="#">u32</a>                |             |
| constructor | <a href="#">MethodSpecV5</a>       |             |
| parents     | <a href="#">StringList</a>         |             |
| hash        | <a href="#">u32</a>                |             |

Named by [CustomTypeDataV5](#).

## ComponentConfig STRUCTURE

Basic runtime configuration for a component

| Name        | Type                        | Description                                 |
|-------------|-----------------------------|---------------------------------------------|
| priority    | <a href="#">Priority</a>    | thread priority                             |
| stackSize   | <a href="#">u32</a>         | thread stack size in bytes, 0 means default |
| group       | <a href="#">u32</a>         | the group where to run the component        |
| cpuAffinity | <a href="#">u64</a>         | CPU affinity mask                           |
| inQueue     | <a href="#">QueueConfig</a> | queuing of inbound information              |
| outQueue    | <a href="#">QueueConfig</a> | queuing of outbound information             |
| sleepPolicy | <a href="#">SleepPolicy</a> | configurable component sleep policy         |

Named by [BuiltComponentParams](#), [LoadedComponentParams](#).

## ComponentInfo STRUCTURE

Basic information of a kernel component

| Name        | Type                      | Description |
|-------------|---------------------------|-------------|
| name        | <a href="#">string</a>    |             |
| description | <a href="#">string</a>    |             |
| buildInfo   | <a href="#">BuildInfo</a> |             |

## CustomTypeSpec STRUCTURE

spec of a custom type

| Name          | Type                           | Description            |
|---------------|--------------------------------|------------------------|
| name          | <a href="#">string</a>         | name of the type       |
| qualifiedName | <a href="#">string</a>         | package-qualified name |
| description   | <a href="#">string</a>         | documentation          |
| data          | <a href="#">CustomTypeData</a> | spec data              |

Named by [sen.components.jsonrpc.TypeLookupResult](#), [CustomTypeSpecList](#).

## CustomTypeSpecV4 STRUCTURE

| Name          | Type                             | Description            |
|---------------|----------------------------------|------------------------|
| name          | <a href="#">string</a>           | name of the type       |
| qualifiedName | <a href="#">string</a>           | package-qualified name |
| description   | <a href="#">string</a>           | documentation          |
| data          | <a href="#">CustomTypeDataV4</a> | spec data              |

Named by [CustomTypeSpecListV4](#).

## CustomTypeSpecV5 STRUCTURE

| Name          | Type                             | Description            |
|---------------|----------------------------------|------------------------|
| name          | <a href="#">string</a>           | name of the type       |
| qualifiedName | <a href="#">string</a>           | package-qualified name |
| description   | <a href="#">string</a>           | documentation          |
| data          | <a href="#">CustomTypeDataV5</a> | spec data              |

Named by [CustomTypeSpecListV5](#).

## EnumTypeSpec STRUCTURE

spec of an enum type

| Name        | Type                               | Description                  |
|-------------|------------------------------------|------------------------------|
| enums       | <a href="#">EnumeratorSpecList</a> | enumerators                  |
| storageType | <a href="#">IntegralType</a>       | type used to store the value |

Named by [CustomTypeData](#).

## EnumTypeSpecV4 STRUCTURE

| Name        | Type                                 | Description                  |
|-------------|--------------------------------------|------------------------------|
| enums       | <a href="#">EnumeratorSpecListV4</a> | enumerators                  |
| storageType | <a href="#">IntegralType</a>         | type used to store the value |

Named by [CustomTypeDataV4](#).

## EnumTypeSpecV5 STRUCTURE

| Name        | Type                                 | Description                  |
|-------------|--------------------------------------|------------------------------|
| enums       | <a href="#">EnumeratorSpecListV5</a> | enumerators                  |
| storageType | <a href="#">IntegralType</a>         | type used to store the value |
| hash        | <a href="#">u32</a>                  | unique hash                  |

Named by [CustomTypeDataV5](#).

## EnumeratorSpec STRUCTURE

an enumerator

| Name        | Type                   | Description            |
|-------------|------------------------|------------------------|
| name        | <a href="#">string</a> | enumerator name        |
| key         | <a href="#">u32</a>    | enumerator value       |
| description | <a href="#">string</a> | enumerator description |

Named by [EnumeratorSpecList](#).

## EnumeratorSpecV4 STRUCTURE

| Name | Type                   | Description      |
|------|------------------------|------------------|
| name | <a href="#">string</a> | enumerator name  |
| key  | <a href="#">u64</a>    | enumerator value |

Named by [EnumeratorSpecListV4](#).

## EnumeratorSpecV5 STRUCTURE

| Name | Type                   | Description      |
|------|------------------------|------------------|
| name | <a href="#">string</a> | enumerator name  |
| key  | <a href="#">u64</a>    | enumerator value |
| hash | <a href="#">u32</a>    | unique hash      |

Named by [EnumeratorSpecListV5](#).

## EnvVar STRUCTURE

Information about an environment variable

| Name  | Type                   | Description           |
|-------|------------------------|-----------------------|
| name  | <a href="#">string</a> | name of the variable  |
| value | <a href="#">string</a> | value of the variable |

Named by [EnvironmentVarList](#).

## ErrorData STRUCTURE

Information related to the error

| Name          | Type                              | Description                                        |
|---------------|-----------------------------------|----------------------------------------------------|
| time          | <a href="#">sen.TimeStamp</a>     | time of the error                                  |
| errorMessage  | <a href="#">StringList</a>        | message presented to the user                      |
| exceptionData | <a href="#">UncaughtException</a> | information about the unhandled exception (if any) |
| signalData    | <a href="#">SignalData</a>        | information about the received signal (if any)     |

Named by [ErrorReport](#).

## ErrorReport STRUCTURE

Information about a detected error

| Name        | Type                        | Description                        |
|-------------|-----------------------------|------------------------------------|
| senData     | <a href="#">SenData</a>     | information related to Sen         |
| processData | <a href="#">ProcessData</a> | information related to the process |
| errorData   | <a href="#">ErrorData</a>   | information related to the error   |

## EventSpec STRUCTURE

spec for an event

| Name          | Type                              | Description      |
|---------------|-----------------------------------|------------------|
| name          | <a href="#">string</a>            | name of the type |
| description   | <a href="#">string</a>            | documentation    |
| args          | <a href="#">ArgSpecList</a>       | arguments        |
| transportMode | <a href="#">TransportModeSpec</a> | transport mode   |

Named by [EventSpecList](#).

## EventSpecV4 STRUCTURE

| Name          | Type              | Description              |
|---------------|-------------------|--------------------------|
| memberHash    | u32               | hash of the class member |
| name          | string            | name of the type         |
| description   | string            | documentation            |
| args          | ArgSpecListV4     | arguments                |
| transportMode | TransportModeSpec | transport mode           |

Named by [EventSpecListV4](#).

## EventSpecV5 STRUCTURE

| Name          | Type              | Description              |
|---------------|-------------------|--------------------------|
| memberHash    | u32               | hash of the class member |
| name          | string            | name of the type         |
| description   | string            | documentation            |
| args          | ArgSpecListV5     | arguments                |
| transportMode | TransportModeSpec | transport mode           |
| hash          | u32               | unique hash              |

Named by [EventSpecListV5](#).

## ExecError STRUCTURE

Holds details of an execution error

| Name        | Type          | Description                |
|-------------|---------------|----------------------------|
| category    | ErrorCategory | general problem            |
| explanation | string        | human-readable explanation |

## KernelParams STRUCTURE

Parameters to configure kernel execution

| Name                | Type              | Description                                                                                                                                                                                                                                              |
|---------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| runMode             | RunMode           | how to run the kernel                                                                                                                                                                                                                                    |
| appName             | string            | optional name of the application                                                                                                                                                                                                                         |
| bus                 | string            | where to publish the kernel objects (defaults to local.kernel)                                                                                                                                                                                           |
| clockBus            | string            | where to publish the virtual clock (if any), defaults to bus                                                                                                                                                                                             |
| clockName           | string            | the name of the virtual clock (if any), defaults to 'clock'                                                                                                                                                                                              |
| clockMaster         | bool              | if time is virtualized, publish a master clock to clockBus                                                                                                                                                                                               |
| logConfig           | log.Config        | logging configuration                                                                                                                                                                                                                                    |
| crashReportDir      | string            | where to store crash reports (defaults to the temp dir)                                                                                                                                                                                                  |
| crashReportDisabled | bool              | if true, no reports are generated                                                                                                                                                                                                                        |
| lockMemoryPages     | bool              | keep process pages memory-resident                                                                                                                                                                                                                       |
| sleepPolicy         | SleepPolicy       | configurable sleep policy of the kernel component                                                                                                                                                                                                        |
| compatibilityMode   | CompatibilityMode | what to do with a remote type that does not match                                                                                                                                                                                                        |
| limitCpuIdleLatency | bool              | This struct is reachable over the wire through Kernel.getConfig() and is embedded in SenData, and the encoding is positional, so adding a field here shifts what follows it either way. keep the CPUs out of the deep idle states they are slow to leave |

Named by [KernelApi](#), [SenData](#).

## LoadedComponentParams STRUCTURE

Parameters related to a component that will be loaded by the kernel

| Name   | Type                            | Description                        |
|--------|---------------------------------|------------------------------------|
| name   | <a href="#">string</a>          | the name of the component          |
| config | <a href="#">ComponentConfig</a> | the configuration of the component |

Named by [LoadedComponentList](#).

## MethodSpec STRUCTURE

spec for methods

| Name             | Type                                 | Description                                   |
|------------------|--------------------------------------|-----------------------------------------------|
| name             | <a href="#">string</a>               | name of the type                              |
| description      | <a href="#">string</a>               | documentation                                 |
| args             | <a href="#">ArgSpecList</a>          | arguments                                     |
| transportMode    | <a href="#">TransportModeSpec</a>    | transport mode                                |
| constness        | <a href="#">MethodConstnessSpec</a>  | if it changes the state of the object         |
| deferred         | <a href="#">bool</a>                 | if the execution of the method is deferred    |
| returnType       | <a href="#">string</a>               | qualified return type name                    |
| propertyRelation | <a href="#">PropertyRelationSpec</a> | property relation of the method               |
| localOnly        | <a href="#">bool</a>                 | true if the method can only be called locally |

Named by [ClassTypeSpec](#), [MethodSpecList](#).

## MethodSpecV4 STRUCTURE

| Name          | Type                                | Description                                 |
|---------------|-------------------------------------|---------------------------------------------|
| memberHash    | <a href="#">u32</a>                 | hash of the class member                    |
| name          | <a href="#">string</a>              | name of the type                            |
| description   | <a href="#">string</a>              | documentation                               |
| args          | <a href="#">ArgSpecListV4</a>       | arguments                                   |
| transportMode | <a href="#">TransportModeSpec</a>   | transport mode                              |
| constness     | <a href="#">MethodConstnessSpec</a> | if it changes the state of the object       |
| deferred      | <a href="#">bool</a>                | true if the method is marked as deferred    |
| returnTypeId  | <a href="#">string</a>              | qualified return type name (empty for none) |

Named by [ClassTypeSpecV4](#), [MethodSpecListV4](#).

## MethodSpecV5 STRUCTURE

| Name          | Type                                | Description                                   |
|---------------|-------------------------------------|-----------------------------------------------|
| memberHash    | <a href="#">u32</a>                 | hash of the class member                      |
| name          | <a href="#">string</a>              | name of the type                              |
| description   | <a href="#">string</a>              | documentation                                 |
| args          | <a href="#">ArgSpecListV5</a>       | arguments                                     |
| transportMode | <a href="#">TransportModeSpec</a>   | transport mode                                |
| constness     | <a href="#">MethodConstnessSpec</a> | if it changes the state of the object         |
| localOnly     | <a href="#">bool</a>                | true if the method can only be called locally |

| Name         | Type                   | Description                                 |
|--------------|------------------------|---------------------------------------------|
| returnTypeId | <a href="#">string</a> | qualified return type name (empty for none) |
| hash         | <a href="#">u32</a>    | unique hash                                 |

Named by [ClassTypeSpecV5](#), [MethodSpecListV5](#).

## OpFinished STRUCTURE

An operation is complete

Named by [OpState](#).

## OpNotFinished STRUCTURE

An operation is not complete

| Name     | Type                         | Description                         |
|----------|------------------------------|-------------------------------------|
| waitHint | <a href="#">sen.Duration</a> | How long to wait until the next try |

Named by [OpState](#).

## OptionalTypeSpec STRUCTURE

data of an optional type

| Name | Type                   | Description         |
|------|------------------------|---------------------|
| type | <a href="#">string</a> | qualified type name |

Named by [CustomTypeData](#).

## OptionalTypeSpecV4 STRUCTURE

| Name | Type                   | Description         |
|------|------------------------|---------------------|
| type | <a href="#">string</a> | qualified type name |

Named by [CustomTypeDataV4](#).

## OptionalTypeSpecV5 STRUCTURE

| Name | Type                   | Description         |
|------|------------------------|---------------------|
| type | <a href="#">string</a> | qualified type name |
| hash | <a href="#">u32</a>    | unique hash         |

Named by [CustomTypeDataV5](#).

## PrecisionSleep STRUCTURE

Be more precise at the expense of some CPU cycles.

| Name                         | Type                         | Description                                |
|------------------------------|------------------------------|--------------------------------------------|
| veryCoarseGrainSleep<br>Time | <a href="#">sen.Duration</a> | First set of sleeps. If 0 defaults to 7ms  |
| coarseGrainSleepTime         | <a href="#">sen.Duration</a> | Second set of sleeps. If 0 defaults to 1ms |

Named by [SleepPolicy](#).

## ProcessData STRUCTURE

Information about the process

| Name        | Type                               | Description                          |
|-------------|------------------------------------|--------------------------------------|
| processInfo | <a href="#">ProcessInfo</a>        | general process information          |
| environment | <a href="#">EnvironmentVarList</a> | list of environment variables        |
| stacktrace  | <a href="#">StringList</a>         | the stack trace (if any)             |
| recentLogs  | <a href="#">StringList</a>         | the last logs emitted by the process |

Named by [ErrorReport](#).

## ProcessInfo STRUCTURE

Basic information about a process

| Name        | Type                    | Description |
|-------------|-------------------------|-------------|
| hostId      | <a href="#">u32</a>     |             |
| processId   | <a href="#">u32</a>     |             |
| sessionId   | <a href="#">u32</a>     |             |
| sessionName | <a href="#">string</a>  |             |
| appName     | <a href="#">string</a>  |             |
| hostName    | <a href="#">string</a>  |             |
| osKind      | <a href="#">OsKind</a>  |             |
| osName      | <a href="#">string</a>  |             |
| cpuArch     | <a href="#">CpuArch</a> |             |

Named by [sen.components.ether.Hello](#), [sen.components.ether.SessionPresenceBeam](#), [ProcessData](#), [Processes](#).

## PropertySpec STRUCTURE

spec of a property

| Name          | Type                                 | Description                         |
|---------------|--------------------------------------|-------------------------------------|
| name          | <a href="#">string</a>               | name of the type                    |
| description   | <a href="#">string</a>               | documentation                       |
| category      | <a href="#">PropertyCategorySpec</a> | category                            |
| type          | <a href="#">string</a>               | qualified type name of the property |
| transportMode | <a href="#">TransportModeSpec</a>    | how to transport the data           |
| tags          | <a href="#">StringList</a>           | user-defined tags                   |
| checkedSet    | <a href="#">bool</a>                 | if setting the property is checked  |

Named by [PropertySpecList](#).

## PropertySpecV4 STRUCTURE

| Name          | Type                                 | Description                         |
|---------------|--------------------------------------|-------------------------------------|
| memberHash    | <a href="#">u32</a>                  | hash of the class member            |
| name          | <a href="#">string</a>               | name of the type                    |
| description   | <a href="#">string</a>               | documentation                       |
| category      | <a href="#">PropertyCategorySpec</a> | category                            |
| type          | <a href="#">string</a>               | qualified type name of the property |
| transportMode | <a href="#">TransportModeSpec</a>    | how to transport the data           |
| tags          | <a href="#">StringList</a>           | user-defined tags                   |

Named by [PropertySpecListV4](#).

## PropertySpecV5 STRUCTURE

| Name          | Type                 | Description                         |
|---------------|----------------------|-------------------------------------|
| memberHash    | u32                  | hash of the class member            |
| name          | string               | name of the type                    |
| description   | string               | documentation                       |
| category      | PropertyCategorySpec | category                            |
| type          | string               | qualified type name of the property |
| transportMode | TransportModeSpec    | how to transport the data           |
| tags          | StringList           | user-defined tags                   |
| hash          | u32                  | unique hash                         |

Named by [PropertySpecListV5](#).

## QuantityTypeSpec STRUCTURE

spec of a quantity type

| Name        | Type        | Description                  |
|-------------|-------------|------------------------------|
| elementType | NumericType | type used to store the value |
| unit        | UnitInfo    | used unit                    |
| minValue    | f64         | discarded if >= maxValue     |
| maxValue    | f64         | discarded if <= minValue     |

Named by [CustomTypeData](#).

## QuantityTypeSpecV4 STRUCTURE

| Name        | Type        | Description                  |
|-------------|-------------|------------------------------|
| numericType | NumericType | type used to store the value |
| unit        | UnitInfo    | used unit                    |
| minValue    | f64         | discarded if >= maxValue     |
| maxValue    | f64         | discarded if <= minValue     |

Named by [CustomTypeDataV4](#).

## QuantityTypeSpecV5 STRUCTURE

| Name        | Type        | Description                  |
|-------------|-------------|------------------------------|
| numericType | NumericType | type used to store the value |
| unit        | UnitInfo    | used unit                    |
| minValue    | f64         | discarded if >= maxValue     |
| maxValue    | f64         | discarded if <= minValue     |
| hash        | u32         | unique hash                  |

Named by [CustomTypeDataV5](#).

## QueueConfig STRUCTURE

| Name           | Type                | Description                       |
|----------------|---------------------|-----------------------------------|
| evictionPolicy | QueueEvictionPolicy | what to do when the queue is full |
| maxSize        | u64                 | 0 means unbounded                 |

Named by [ComponentConfig](#).

## SenData STRUCTURE

Information about Sen

| Name              | Type                                | Description                                      |
|-------------------|-------------------------------------|--------------------------------------------------|
| version           | <a href="#">string</a>              | the Sen version                                  |
| compiler          | <a href="#">string</a>              | the compiler used to build Sen                   |
| debugMode         | <a href="#">bool</a>                | true if it was build in debug mode               |
| buildTime         | <a href="#">string</a>              | the commit's timestamp, not the compiler's clock |
| wordSize          | <a href="#">WordSize</a>            | architecture                                     |
| gitRef            | <a href="#">string</a>              | git ref                                          |
| gitHash           | <a href="#">string</a>              | git hash                                         |
| gitStatus         | <a href="#">GitStatus</a>           | status of the branch when build                  |
| kernelParams      | <a href="#">KernelParams</a>        | Sen kernel parameters                            |
| kernelProtocol    | <a href="#">u32</a>                 | version of the kernel protocol                   |
| transportProtocol | <a href="#">u32</a>                 | version of the transport protocol (if present)   |
| loadedComponents  | <a href="#">LoadedComponentList</a> | components to load                               |
| builtComponents   | <a href="#">BuiltComponentList</a>  | components to build                              |

Named by [ErrorReport](#).

## SequenceTypeSpec STRUCTURE

spec of a sequence type

| Name        | Type                   | Description                           |
|-------------|------------------------|---------------------------------------|
| elementType | <a href="#">string</a> | qualified type name for stored values |
| maxSize     | <a href="#">u64</a>    | discarded when 0                      |
| fixedSize   | <a href="#">bool</a>   |                                       |

Named by [CustomTypeData](#).

## SequenceTypeSpecV4 STRUCTURE

| Name        | Type                   | Description                           |
|-------------|------------------------|---------------------------------------|
| elementType | <a href="#">string</a> | qualified type name for stored values |
| maxSize     | <a href="#">u64</a>    | discarded when 0                      |

Named by [CustomTypeDataV4](#).

## SequenceTypeSpecV5 STRUCTURE

| Name        | Type                   | Description                           |
|-------------|------------------------|---------------------------------------|
| elementType | <a href="#">string</a> | qualified type name for stored values |
| maxSize     | <a href="#">u64</a>    | discarded when 0                      |
| fixedSize   | <a href="#">bool</a>   |                                       |
| hash        | <a href="#">u32</a>    | unique hash                           |

Named by [CustomTypeDataV5](#).

## SignalData STRUCTURE

Information about an unhandled signal

| Name         | Type                   | Description         |
|--------------|------------------------|---------------------|
| signalNumber | <a href="#">i32</a>    | signal number       |
| signalName   | <a href="#">string</a> | human-readable name |

Named by [ErrorData](#).

## StructTypeFieldSpec STRUCTURE

a field of a struct

| Name        | Type                   | Description                            |
|-------------|------------------------|----------------------------------------|
| name        | <a href="#">string</a> | field name                             |
| description | <a href="#">string</a> | documentation                          |
| type        | <a href="#">string</a> | qualified type name held by this field |

Named by [StructTypeFieldSpecList](#).

## StructTypeFieldSpecV4 STRUCTURE

| Name        | Type                   | Description                            |
|-------------|------------------------|----------------------------------------|
| name        | <a href="#">string</a> | field name                             |
| description | <a href="#">string</a> | documentation                          |
| type        | <a href="#">string</a> | qualified type name held by this field |

Named by [StructTypeFieldSpecListV4](#).

## StructTypeFieldSpecV5 STRUCTURE

| Name        | Type                   | Description                            |
|-------------|------------------------|----------------------------------------|
| name        | <a href="#">string</a> | field name                             |
| description | <a href="#">string</a> | documentation                          |
| type        | <a href="#">string</a> | qualified type name held by this field |
| hash        | <a href="#">u32</a>    | unique hash                            |

Named by [StructTypeFieldSpecListV5](#).

## StructTypeSpec STRUCTURE

spec of an struct type

| Name   | Type                                    | Description                                             |
|--------|-----------------------------------------|---------------------------------------------------------|
| fields | <a href="#">StructTypeFieldSpecList</a> | fields                                                  |
| parent | <a href="#">string</a>                  | qualified type name the parent struct, empty means none |

Named by [CustomTypeData](#).

## StructTypeSpecV4 STRUCTURE

| Name   | Type                                      | Description                                             |
|--------|-------------------------------------------|---------------------------------------------------------|
| fields | <a href="#">StructTypeFieldSpecListV4</a> | fields                                                  |
| parent | <a href="#">string</a>                    | qualified type name the parent struct, empty means none |

Named by [CustomTypeDataV4](#).

## StructTypeSpecV5 STRUCTURE

| Name   | Type                                      | Description |
|--------|-------------------------------------------|-------------|
| fields | <a href="#">StructTypeFieldSpecListV5</a> | fields      |

| Name   | Type                   | Description                                             |
|--------|------------------------|---------------------------------------------------------|
| parent | <a href="#">string</a> | qualified type name the parent struct, empty means none |
| hash   | <a href="#">u32</a>    | unique hash                                             |

Named by [CustomTypeDataV5](#).

## SystemSleep STRUCTURE

Use the native system sleep

Named by [SleepPolicy](#).

## UncaughtException STRUCTURE

Information about an unhandled exception

| Name          | Type                          | Description                  |
|---------------|-------------------------------|------------------------------|
| exceptionKind | <a href="#">ExceptionKind</a> | kind of exception            |
| message       | <a href="#">string</a>        | message it provides (if any) |

Named by [ErrorData](#).

## UnitInfo STRUCTURE

Describes a unit

| Name         | Type                    | Description |
|--------------|-------------------------|-------------|
| name         | <a href="#">string</a>  |             |
| abbreviation | <a href="#">string</a>  |             |
| category     | <a href="#">UnitCat</a> |             |

Named by [QuantityTypeSpec](#), [QuantityTypeSpecV4](#), [QuantityTypeSpecV5](#), [UnitList](#).

## VariantTypeFieldSpec STRUCTURE

a field of a variant

| Name        | Type                   | Description                               |
|-------------|------------------------|-------------------------------------------|
| key         | <a href="#">u32</a>    | key/index of this type within the variant |
| description | <a href="#">string</a> | documentation                             |
| type        | <a href="#">string</a> | qualified type name held by this field    |

Named by [VariantTypeFieldSpecList](#).

## VariantTypeFieldSpecV4 STRUCTURE

| Name        | Type                   | Description                               |
|-------------|------------------------|-------------------------------------------|
| key         | <a href="#">u32</a>    | key/index of this type within the variant |
| description | <a href="#">string</a> | documentation                             |
| type        | <a href="#">string</a> | qualified type name held by this field    |

Named by [VariantTypeFieldSpecListV4](#).

## VariantTypeFieldSpecV5 STRUCTURE

| Name        | Type                   | Description                               |
|-------------|------------------------|-------------------------------------------|
| key         | <a href="#">u32</a>    | key/index of this type within the variant |
| description | <a href="#">string</a> | documentation                             |
| type        | <a href="#">string</a> | qualified type name held by this field    |

| Name | Type | Description |
|------|------|-------------|
| hash | u32  | unique hash |

Named by [VariantTypeFieldSpecListV5](#).

## VariantTypeSpec STRUCTURE

spec of an struct type

| Name   | Type                                     | Description |
|--------|------------------------------------------|-------------|
| fields | <a href="#">VariantTypeFieldSpecList</a> |             |

Named by [CustomTypeData](#).

## VariantTypeSpecV4 STRUCTURE

| Name   | Type                                       | Description |
|--------|--------------------------------------------|-------------|
| fields | <a href="#">VariantTypeFieldSpecListV4</a> |             |

Named by [CustomTypeDataV4](#).

## VariantTypeSpecV5 STRUCTURE

| Name   | Type                                       | Description |
|--------|--------------------------------------------|-------------|
| fields | <a href="#">VariantTypeFieldSpecListV5</a> |             |
| hash   | u32                                        |             |

Named by [CustomTypeDataV5](#).

### Enumerations

|                                      |                                                                                 |    |
|--------------------------------------|---------------------------------------------------------------------------------|----|
| <a href="#">BasicType</a>            | built-in others                                                                 | 59 |
| <a href="#">CompatibilityMode</a>    | What to do when a type published by another kernel does not match the local one | 60 |
| <a href="#">ComponentState</a>       | State of a kernel component                                                     | 60 |
| <a href="#">CpuArch</a>              | Type of CPU                                                                     | 60 |
| <a href="#">ErrorCategory</a>        | Category of a problem during component execution                                | 60 |
| <a href="#">ExceptionKind</a>        | Types of exceptions that can occur.                                             | 61 |
| <a href="#">GitStatus</a>            | Status of git                                                                   | 61 |
| <a href="#">IntegralType</a>         | built-in integrals                                                              | 61 |
| <a href="#">MethodConstnessSpec</a>  | method constness                                                                | 61 |
| <a href="#">OsKind</a>               | Type of operating system                                                        | 61 |
| <a href="#">Priority</a>             | Thread execution priorities                                                     | 62 |
| <a href="#">PropertyCategorySpec</a> | how a property is seen by others                                                | 62 |
| <a href="#">PropertyRelationSpec</a> | method property relation                                                        | 62 |
| <a href="#">QueueEvictionPolicy</a>  | Component queue eviction policy                                                 | 62 |
| <a href="#">RealType</a>             | built-in reals                                                                  | 62 |
| <a href="#">RunMode</a>              | Run modes for the kernel                                                        | 62 |
| <a href="#">ThreadCreateErr</a>      | What can go wrong when creating a thread                                        | 63 |
| <a href="#">TransportModeSpec</a>    | how to transport information                                                    | 63 |
| <a href="#">UnitCat</a>              | Type of measurement                                                             | 63 |
| <a href="#">WordSize</a>             | Native processor word size                                                      | 63 |

## BasicType ENUMERATION

built-in others

Held as u8.

|              |   |               |   |
|--------------|---|---------------|---|
| booleanType  | 0 | timestampType | 3 |
| stringType   | 1 |               |   |
| durationType | 2 |               |   |

Named by [BuiltInType](#).

---

## CompatibilityMode ENUMERATION

What to do when a type published by another kernel does not match the local one

Held as u8.

|          |   |
|----------|---|
| relaxed  | 0 |
| strict   | 1 |
| disabled | 2 |

Named by [KernelParams](#).

---

## ComponentState ENUMERATION

State of a kernel component

Held as u8.

|              |    |
|--------------|----|
| unloaded     | 0  |
| preloaded    | 1  |
| loading      | 2  |
| loaded       | 3  |
| initializing | 4  |
| initialized  | 5  |
| running      | 6  |
| stopping     | 7  |
| stopped      | 8  |
| unloading    | 9  |
| error        | 10 |

---

## CpuArch ENUMERATION

Type of CPU

Held as u8.

|           |    |
|-----------|----|
| x86       | 0  |
| x64       | 1  |
| arm2      | 2  |
| arm3      | 3  |
| arm4T     | 4  |
| arm5      | 5  |
| arm6T2    | 6  |
| arm6      | 7  |
| arm7      | 8  |
| arm7a     | 9  |
| arm7r     | 10 |
| arm7s     | 11 |
| arm64     | 12 |
| mips      | 13 |
| superH    | 14 |
| ppc       | 15 |
| ppc64     | 16 |
| sparc     | 17 |
| m68k      | 18 |
| otherArch | 19 |

Named by [ProcessInfo](#).

---

## ErrorCategory ENUMERATION

Category of a problem during component execution

Held as u8.

|                    |   |
|--------------------|---|
| runtimeError       | 0 |
| logicError         | 1 |
| expectationsNotMet | 2 |
| ioError            | 3 |
| other              | 4 |

Named by [ExecError](#).

---

## ExceptionKind ENUMERATION

Types of exceptions that can occur.

Held as u8.

|          |   |
|----------|---|
| runtime  | 0 |
| logic    | 1 |
| standard | 2 |
| unknown  | 3 |

Named by [UncaughtException](#).

---

## GitStatus ENUMERATION

Status of git

Held as u8.

|          |   |
|----------|---|
| clean    | 0 |
| modified | 1 |
| unknown  | 2 |

Named by [BuildInfo](#), [SenData](#).

---

## IntegralType ENUMERATION

built-in integrals

Held as u8.

|            |   |
|------------|---|
| uint8Type  | 0 |
| int16Type  | 1 |
| uint16Type | 2 |
| int32Type  | 3 |
| uint32Type | 4 |
| int64Type  | 5 |
| uint64Type | 6 |

Named by [EnumTypeSpec](#), [EnumTypeSpecV4](#), [EnumTypeSpecV5](#), [NumericType](#).

---

## MethodConstnessSpec ENUMERATION

method constness

Held as u8.

|             |   |
|-------------|---|
| constant    | 0 |
| nonConstant | 1 |

Named by [MethodSpec](#), [MethodSpecV4](#), [MethodSpecV5](#).

---

## OsKind ENUMERATION

Type of operating system

Held as u8.

|           |   |
|-----------|---|
| windowsOs | 0 |
| linuxOs   | 1 |
| androidOs | 2 |
| appleOs   | 3 |
| unixOs    | 4 |
| posixOs   | 5 |
| otherOs   | 6 |

Named by [ProcessInfo](#).

## Priority ENUMERATION

Thread execution priorities

Held as u32.

|            |   |
|------------|---|
| lowest     | 0 |
| nominalMin | 1 |
| nominalMax | 2 |
| highest    | 3 |

Named by [ComponentConfig](#).

## PropertyCategorySpec ENUMERATION

how a property is seen by others

Held as u8.

|           |   |
|-----------|---|
| staticRW  | 0 |
| staticRO  | 1 |
| dynamicRW | 2 |
| dynamicRO | 3 |

Named by [PropertySpec](#), [PropertySpecV4](#), [PropertySpecV5](#).

## PropertyRelationSpec ENUMERATION

method property relation

Held as u8.

|                    |   |
|--------------------|---|
| nonPropertyRelated | 0 |
| propertyGetter     | 1 |
| propertySetter     | 2 |

Named by [MethodSpec](#).

## QueueEvictionPolicy ENUMERATION

Component queue eviction policy

Held as u32.

|            |   |
|------------|---|
| dropOldest | 0 |
| dropNewest | 1 |

Named by [QueueConfig](#).

## RealType ENUMERATION

built-in reals

Held as u8.

|             |   |
|-------------|---|
| float32Type | 0 |
| float64Type | 1 |

Named by [NumericType](#).

## RunMode ENUMERATION

Run modes for the kernel

Held as u8.

|                    |   |
|--------------------|---|
| realTime           | 0 |
| virtualTime        | 1 |
| virtualTimeRunning | 2 |
| startAndStop       | 3 |

Named by [KernelParams](#).

## ThreadCreateErr ENUMERATION

What can go wrong when creating a thread

Held as u32.

|                               |   |
|-------------------------------|---|
| schedulerAlreadyStarted       | 0 |
| invalidStackSize              | 1 |
| invalidThreadFunction         | 2 |
| invalidThreadFunctionArgument | 3 |
| invalidAffinity               | 4 |
| tooManyThreads                | 5 |
| internalOsError               | 6 |

## TransportModeSpec ENUMERATION

how to transport information

Held as u8.

|           |   |
|-----------|---|
| unicast   | 0 |
| multicast | 1 |
| confirmed | 2 |

Named by [EventSpec](#), [EventSpecV4](#), [EventSpecV5](#), [MethodSpec](#), [MethodSpecV4](#), [MethodSpecV5](#), [PropertySpec](#), [PropertySpecV4](#), [PropertySpecV5](#).

## UnitCat ENUMERATION

Type of measurement

Held as u8.

|                     |    |
|---------------------|----|
| length              | 0  |
| mass                | 1  |
| time                | 2  |
| angle               | 3  |
| temperature         | 4  |
| frequency           | 5  |
| velocity            | 6  |
| angularVelocity     | 7  |
| acceleration        | 8  |
| angularAcceleration | 9  |
| density             | 10 |
| pressure            | 11 |
| area                | 12 |
| force               | 13 |
| torque              | 14 |

Named by [UnitInfo](#).

## WordSize ENUMERATION

Native processor word size

Held as u32.

|        |   |
|--------|---|
| bits32 | 0 |
| bits64 | 1 |

Named by [BuildInfo](#), [SenData](#).

### Variant records

|                                  |                            |    |
|----------------------------------|----------------------------|----|
| <a href="#">BuiltInType</a>      | all built-in types         | 64 |
| <a href="#">CustomTypeData</a>   | all custom types           | 64 |
| <a href="#">CustomTypeDataV4</a> |                            | 64 |
| <a href="#">CustomTypeDataV5</a> |                            | 64 |
| <a href="#">NumericType</a>      | all built-in numeric types | 64 |
| <a href="#">OpState</a>          | The status of an operation | 65 |
| <a href="#">SleepPolicy</a>      | Component sleep policy     | 65 |

## BuiltInType VARIANT

all built-in types

| Type                        | Description |
|-----------------------------|-------------|
| <a href="#">NumericType</a> |             |
| <a href="#">BasicType</a>   |             |

## CustomTypeData VARIANT

all custom types

| Type                             | Description |
|----------------------------------|-------------|
| <a href="#">EnumTypeSpec</a>     |             |
| <a href="#">QuantityTypeSpec</a> |             |
| <a href="#">SequenceTypeSpec</a> |             |
| <a href="#">StructTypeSpec</a>   |             |
| <a href="#">VariantTypeSpec</a>  |             |
| <a href="#">AliasTypeSpec</a>    |             |
| <a href="#">OptionalTypeSpec</a> |             |
| <a href="#">ClassTypeSpec</a>    |             |

Named by [CustomTypeSpec](#).

## CustomTypeDataV4 VARIANT

| Type                               | Description |
|------------------------------------|-------------|
| <a href="#">EnumTypeSpecV4</a>     |             |
| <a href="#">QuantityTypeSpecV4</a> |             |
| <a href="#">SequenceTypeSpecV4</a> |             |
| <a href="#">StructTypeSpecV4</a>   |             |
| <a href="#">VariantTypeSpecV4</a>  |             |
| <a href="#">AliasTypeSpecV4</a>    |             |
| <a href="#">OptionalTypeSpecV4</a> |             |
| <a href="#">ClassTypeSpecV4</a>    |             |

Named by [CustomTypeSpecV4](#).

## CustomTypeDataV5 VARIANT

| Type                               | Description |
|------------------------------------|-------------|
| <a href="#">EnumTypeSpecV5</a>     |             |
| <a href="#">QuantityTypeSpecV5</a> |             |
| <a href="#">SequenceTypeSpecV5</a> |             |
| <a href="#">StructTypeSpecV5</a>   |             |
| <a href="#">VariantTypeSpecV5</a>  |             |
| <a href="#">AliasTypeSpecV5</a>    |             |
| <a href="#">OptionalTypeSpecV5</a> |             |
| <a href="#">ClassTypeSpecV5</a>    |             |

Named by [CustomTypeSpecV5](#).

## NumericType VARIANT

all built-in numeric types

| Type                         | Description |
|------------------------------|-------------|
| <a href="#">IntegralType</a> |             |
| <a href="#">RealType</a>     |             |

Named by [BuiltInType](#), [QuantityTypeSpec](#), [QuantityTypeSpecV4](#), [QuantityTypeSpecV5](#).

## OpState VARIANT

The status of an operation

| Type                          | Description |
|-------------------------------|-------------|
| <a href="#">OpFinished</a>    | done        |
| <a href="#">OpNotFinished</a> | not done    |

## SleepPolicy VARIANT

Component sleep policy

| Type                           | Description |
|--------------------------------|-------------|
| <a href="#">PrecisionSleep</a> |             |
| <a href="#">SystemSleep</a>    |             |

Named by [ComponentConfig](#), [KernelParams](#).

### Arrays

|                                            |                                                   |    |
|--------------------------------------------|---------------------------------------------------|----|
| <a href="#">ArgSpecList</a>                | list of argument specs                            | 65 |
| <a href="#">ArgSpecListV4</a>              |                                                   | 66 |
| <a href="#">ArgSpecListV5</a>              |                                                   | 66 |
| <a href="#">Buffer</a>                     | A general-purpose unbounded buffer                | 66 |
| <a href="#">BuiltComponentList</a>         |                                                   | 66 |
| <a href="#">CustomTypeSpecList</a>         | list of custom types                              | 66 |
| <a href="#">CustomTypeSpecListV4</a>       |                                                   | 66 |
| <a href="#">CustomTypeSpecListV5</a>       |                                                   | 66 |
| <a href="#">EnumeratorSpecList</a>         | list of enumerators                               | 66 |
| <a href="#">EnumeratorSpecListV4</a>       |                                                   | 66 |
| <a href="#">EnumeratorSpecListV5</a>       |                                                   | 67 |
| <a href="#">EnvironmentVarList</a>         |                                                   | 67 |
| <a href="#">EventSpecList</a>              | list of event specs                               | 67 |
| <a href="#">EventSpecListV4</a>            |                                                   | 67 |
| <a href="#">EventSpecListV5</a>            |                                                   | 67 |
| <a href="#">LoadedComponentList</a>        |                                                   | 67 |
| <a href="#">MethodSpecList</a>             | list of method specs                              | 67 |
| <a href="#">MethodSpecListV4</a>           |                                                   | 67 |
| <a href="#">MethodSpecListV5</a>           |                                                   | 67 |
| <a href="#">Processes</a>                  | Unbounded sequence of ProcessInfo                 | 68 |
| <a href="#">PropertySpecList</a>           | list of property specs                            | 68 |
| <a href="#">PropertySpecListV4</a>         |                                                   | 68 |
| <a href="#">PropertySpecListV5</a>         |                                                   | 68 |
| <a href="#">StringList</a>                 | Unbounded sequence of strings for general purpose | 68 |
| <a href="#">StructTypeFieldSpecList</a>    | list of struct fields                             | 68 |
| <a href="#">StructTypeFieldSpecListV4</a>  |                                                   | 68 |
| <a href="#">StructTypeFieldSpecListV5</a>  |                                                   | 68 |
| <a href="#">U32List</a>                    |                                                   | 68 |
| <a href="#">UnitList</a>                   | Unbounded sequence of unit information structs    | 69 |
| <a href="#">VariantTypeFieldSpecList</a>   | list of variant fields                            | 69 |
| <a href="#">VariantTypeFieldSpecListV4</a> |                                                   | 69 |
| <a href="#">VariantTypeFieldSpecListV5</a> |                                                   | 69 |

## ArgSpecList SEQUENCE

list of argument specs

|                                                                   |
|-------------------------------------------------------------------|
| <code>sequence&lt;<a href="#">ArgSpec</a>&gt; ArgSpecList;</code> |
|-------------------------------------------------------------------|

Named by [EventSpec](#), [MethodSpec](#).

---

## ArgSpecListV4 SEQUENCE

```
sequence<ArgSpecV4> ArgSpecListV4;
```

Named by [EventSpecV4](#), [MethodSpecV4](#).

---

## ArgSpecListV5 SEQUENCE

```
sequence<ArgSpecV5> ArgSpecListV5;
```

Named by [EventSpecV5](#), [MethodSpecV5](#).

---

## Buffer SEQUENCE

A general-purpose unbounded buffer

```
sequence<u8> Buffer;
```

Named by [sen.components.jsonrpc.StaticFile](#).

---

## BuiltComponentList SEQUENCE

```
sequence<BuiltComponentParams> BuiltComponentList;
```

Named by [SenData](#).

---

## CustomTypeSpecList SEQUENCE

list of custom types

```
sequence<CustomTypeSpec> CustomTypeSpecList;
```

Named by [sen.components.jsonrpc.JsonRpcServer](#).

---

## CustomTypeSpecListV4 SEQUENCE

```
sequence<CustomTypeSpecV4> CustomTypeSpecListV4;
```

---

## CustomTypeSpecListV5 SEQUENCE

```
sequence<CustomTypeSpecV5> CustomTypeSpecListV5;
```

---

## EnumeratorSpecList SEQUENCE

list of enumerators

```
sequence<EnumeratorSpec> EnumeratorSpecList;
```

Named by [EnumTypeSpec](#).

---

## EnumeratorSpecListV4 SEQUENCE

```
sequence<EnumeratorSpecV4> EnumeratorSpecListV4;
```

Named by [EnumTypeSpecV4](#).

---

## EnumeratorSpecListV5 SEQUENCE

```
sequence<EnumeratorSpecV5> EnumeratorSpecListV5;
```

Named by [EnumTypeSpecV5](#).

---

## EnvironmentVarList SEQUENCE

```
sequence<EnvVar> EnvironmentVarList;
```

Named by [ProcessData](#).

---

## EventSpecList SEQUENCE

list of event specs

```
sequence<EventSpec> EventSpecList;
```

Named by [ClassTypeSpec](#).

---

## EventSpecListV4 SEQUENCE

```
sequence<EventSpecV4> EventSpecListV4;
```

Named by [ClassTypeSpecV4](#).

---

## EventSpecListV5 SEQUENCE

```
sequence<EventSpecV5> EventSpecListV5;
```

Named by [ClassTypeSpecV5](#).

---

## LoadedComponentList SEQUENCE

```
sequence<LoadedComponentParams> LoadedComponentList;
```

Named by [SenData](#).

---

## MethodSpecList SEQUENCE

list of method specs

```
sequence<MethodSpec> MethodSpecList;
```

Named by [ClassTypeSpec](#).

---

## MethodSpecListV4 SEQUENCE

```
sequence<MethodSpecV4> MethodSpecListV4;
```

Named by [ClassTypeSpecV4](#).

---

## MethodSpecListV5 SEQUENCE

```
sequence<MethodSpecV5> MethodSpecListV5;
```

Named by [ClassTypeSpecV5](#).

---

## Processes SEQUENCE

Unbounded sequence of ProcessInfo

```
sequence<ProcessInfo> Processes;
```

## PropertySpecList SEQUENCE

list of property specs

```
sequence<PropertySpec> PropertySpecList;
```

Named by [ClassTypeSpec](#).

## PropertySpecListV4 SEQUENCE

```
sequence<PropertySpecV4> PropertySpecListV4;
```

Named by [ClassTypeSpecV4](#).

## PropertySpecListV5 SEQUENCE

```
sequence<PropertySpecV5> PropertySpecListV5;
```

Named by [ClassTypeSpecV5](#).

## StringList SEQUENCE

Unbounded sequence of strings for general purpose

```
sequence<string> StringList;
```

Named by [sen.components.jsonrpc.JsonRpcServer](#), [sen.components.jsonrpc.NamedSelection](#), [sen.components.jsonrpc.SessionInfo](#), [BuiltComponentParams](#), [ClassTypeSpec](#), [ClassTypeSpecV4](#), [ClassTypeSpecV5](#), [ErrorData](#), [KernelApi](#), [ProcessData](#), [PropertySpec](#), [PropertySpecV4](#), [PropertySpecV5](#).

## StructTypeFieldSpecList SEQUENCE

list of struct fields

```
sequence<StructTypeFieldSpec> StructTypeFieldSpecList;
```

Named by [StructTypeSpec](#).

## StructTypeFieldSpecListV4 SEQUENCE

```
sequence<StructTypeFieldSpecV4> StructTypeFieldSpecListV4;
```

Named by [StructTypeSpecV4](#).

## StructTypeFieldSpecListV5 SEQUENCE

```
sequence<StructTypeFieldSpecV5> StructTypeFieldSpecListV5;
```

Named by [StructTypeSpecV5](#).

## U32List SEQUENCE

```
sequence<u32> U32List;
```

UnitList

SEQUENCE

Unbounded sequence of unit information structs

```
sequence<UnitInfo> UnitList;
```

Named by [KernelApi](#).

VariantTypeFieldSpecList

SEQUENCE

list of variant fields

```
sequence<VariantTypeFieldSpec> VariantTypeFieldSpecList;
```

Named by [VariantTypeSpec](#).

VariantTypeFieldSpecListV4

SEQUENCE

```
sequence<VariantTypeFieldSpecV4> VariantTypeFieldSpecListV4;
```

Named by [VariantTypeSpecV4](#).

VariantTypeFieldSpecListV5

SEQUENCE

```
sequence<VariantTypeFieldSpecV5> VariantTypeFieldSpecListV5;
```

Named by [VariantTypeSpecV5](#).

Optionals

|                                        |                          |    |
|----------------------------------------|--------------------------|----|
| <a href="#">MaybeException</a>         |                          | 69 |
| <a href="#">MaybeF64</a>               | optional value for a f64 | 69 |
| <a href="#">MaybeSignal</a>            |                          | 69 |
| <a href="#">MaybeStacktrace</a>        |                          | 69 |
| <a href="#">MaybeTransportProtocol</a> |                          | 69 |
| <a href="#">MaybeU64</a>               | optional value for a u64 | 70 |

MaybeException

OPTIONAL

```
optional<UncaughtException> MaybeException;
```

MaybeF64

OPTIONAL

optional value for a f64

```
optional<f64> MaybeF64;
```

MaybeSignal

OPTIONAL

```
optional<SignalData> MaybeSignal;
```

MaybeStacktrace

OPTIONAL

```
optional<StringList> MaybeStacktrace;
```

MaybeTransportProtocol

OPTIONAL

```
optional<u32> MaybeTransportProtocol;
```

## MaybeU64 OPTIONAL

optional value for a u64

```
optional<u64> MaybeU64;
```

# log

16 types.

## Fixed records

|                              |                                               |    |
|------------------------------|-----------------------------------------------|----|
| <a href="#">BasicFile</a>    | Basic file sink                               | 71 |
| <a href="#">ColorStderr</a>  | Colored stderr sink                           | 71 |
| <a href="#">ColorStdout</a>  | Colored stdout sink                           | 71 |
| <a href="#">Config</a>       | Overall component configuration               | 71 |
| <a href="#">LoggerConfig</a> | Configuration used for a given spdlog logger. | 71 |
| <a href="#">Null</a>         | Single-threaded null sink                     | 72 |
| <a href="#">RotatingFile</a> | Rotating file sink                            | 72 |
| <a href="#">SinkConfig</a>   | Configuration used for a given spdlog Sink.   | 72 |
| <a href="#">Stderr</a>       | stderr sink                                   | 72 |
| <a href="#">Stdout</a>       | stdout sink                                   | 72 |
| <a href="#">Syslog</a>       | Syslog sink (only in linux)                   | 72 |

## BasicFile STRUCTURE

Basic file sink

| Name            | Type                   | Description                                   |
|-----------------|------------------------|-----------------------------------------------|
| fileName        | <a href="#">string</a> | file to log to                                |
| truncate        | <a href="#">bool</a>   | defaults to false                             |
| createParentDir | <a href="#">bool</a>   | true if we should create the parent dir first |

Named by [SinkType](#).

## ColorStderr STRUCTURE

Colored stderr sink

Named by [SinkType](#).

## ColorStdout STRUCTURE

Colored stdout sink

Named by [SinkType](#).

## Config STRUCTURE

Overall component configuration

| Name      | Type                             | Description                          |
|-----------|----------------------------------|--------------------------------------|
| pattern   | <a href="#">string</a>           | default format (empty means default) |
| level     | <a href="#">LogLevel</a>         | default global level for all         |
| sinks     | <a href="#">SinkConfigList</a>   | sinks to configure                   |
| loggers   | <a href="#">LoggerConfigList</a> | loggers to configure                 |
| backtrace | <a href="#">bool</a>             | log backtrace is printed on failure  |

Named by [sen.kernel.KernelParams](#).

## LoggerConfig STRUCTURE

Configuration used for a given spdlog logger.

| Name  | Type                         | Description                                |
|-------|------------------------------|--------------------------------------------|
| name  | <a href="#">string</a>       | name of the logger                         |
| sinks | <a href="#">SinkNameList</a> | name of the sinks to connect the logger to |
| level | <a href="#">LogLevel</a>     | default log level for this logger          |

| Name    | Type                   | Description                                                                                                                                                                                                                                               |
|---------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pattern | <a href="#">string</a> | Formatting for the sinks in this logger. Does nothing if empty. Note that this replaces the format of the associated sinks, and this could affect other loggers using those sinks. The config is applied following the order of the Config.loggers field. |

Named by [LoggerConfigList](#).

## Null STRUCTURE

Single-threaded null sink

Named by [SinkType](#).

## RotatingFile STRUCTURE

Rotating file sink

| Name         | Type                   | Description          |
|--------------|------------------------|----------------------|
| baseFileName | <a href="#">string</a> |                      |
| maxSize      | <a href="#">u32</a>    | 0 means use defaults |
| maxFiles     | <a href="#">u32</a>    | 0 means use defaults |

Named by [SinkType](#).

## SinkConfig STRUCTURE

Configuration used for a given spdlog Sink.

| Name           | Type                     | Description                                                |
|----------------|--------------------------|------------------------------------------------------------|
| name           | <a href="#">string</a>   | name of the sink                                           |
| singleThreaded | <a href="#">bool</a>     | if true, single-threaded (defaults to false)               |
| pattern        | <a href="#">string</a>   | the pattern used for the formatting (empty means default). |
| level          | <a href="#">LogLevel</a> | default log level for this sink                            |
| config         | <a href="#">SinkType</a> | the type-specific configuration parameters                 |

Named by [SinkConfigList](#).

## Stderr STRUCTURE

stderr sink

Named by [SinkType](#).

## Stdout STRUCTURE

stdout sink

Named by [SinkType](#).

## Syslog STRUCTURE

Syslog sink (only in linux)

| Name             | Type                   | Description |
|------------------|------------------------|-------------|
| identity         | <a href="#">string</a> |             |
| option           | <a href="#">u32</a>    |             |
| enableFormatting | <a href="#">bool</a>   |             |

Named by [SinkType](#).

Enumerations

[LogLevel](#)

Severity/importance of a log message.

73

LogLevel ENUMERATION

Severity/importance of a log message.

Held as u8.

|          |   |
|----------|---|
| warn     | 0 |
| err      | 1 |
| critical | 2 |
| trace    | 3 |
| debug    | 4 |
| info     | 5 |
| off      | 6 |

Named by [sen.components.logmaster.LogMaster](#), [sen.components.logmaster.Logger](#), [Config](#), [LoggerConfig](#), [SinkConfig](#).

Variant records

[SinkType](#)

The type-specific configuration of the used spdlog Sink.

73

SinkType VARIANT

The type-specific configuration of the used spdlog Sink.

| Type                         | Description |
|------------------------------|-------------|
| <a href="#">ColorStdout</a>  |             |
| <a href="#">ColorStderr</a>  |             |
| <a href="#">Stdout</a>       |             |
| <a href="#">Stderr</a>       |             |
| <a href="#">BasicFile</a>    |             |
| <a href="#">RotatingFile</a> |             |
| <a href="#">Null</a>         |             |
| <a href="#">Syslog</a>       |             |

Named by [SinkConfig](#).

Arrays

[LoggerConfigList](#)

73

[SinkConfigList](#)

73

[SinkNameList](#)

73

LoggerConfigList SEQUENCE

```
sequence<LoggerConfig> LoggerConfigList;
```

Named by [Config](#).

SinkConfigList SEQUENCE

```
sequence<SinkConfig> SinkConfigList;
```

Named by [Config](#).

SinkNameList SEQUENCE

```
sequence<string> SinkNameList;
```

Named by [LoggerConfig](#).

## Built-in types

The types the language provides. Every one is written the same way on the wire wherever it appears.

| Type                       | Meaning                      |
|----------------------------|------------------------------|
| <code>bool</code>          | classic boolean              |
| <code>f32</code>           | 32 bit floating point number |
| <code>f64</code>           | 64 bit floating point number |
| <code>i32</code>           | 32 bit signed integer        |
| <code>sen.Duration</code>  | a time duration              |
| <code>sen.TimeStamp</code> | a point in time              |
| <code>string</code>        | unbounded string             |
| <code>u16</code>           | 16 bit unsigned integer      |
| <code>u32</code>           | 32 bit unsigned integer      |
| <code>u64</code>           | 64 bit unsigned integer      |
| <code>u8</code>            | 8 bit unsigned integer       |

|                                      |    |                                        |    |                               |    |
|--------------------------------------|----|----------------------------------------|----|-------------------------------|----|
| components.ether.BusConfig           | 7  | components.logmaster.Config            | 26 | kernel.BasicType              | 59 |
| components.ether.BusJoined           | 7  | components.logmaster.LogMaster         | 26 | kernel.Buffer                 | 66 |
| components.ether.BusLeft             | 7  | components.logmaster.Logger            | 26 | kernel.BuildInfo              | 47 |
| components.ether.ByteRange           | 8  | components.py.Configuration            | 27 | kernel.BuiltComponentList     | 66 |
| components.ether.Configuration       | 8  | components.py.PythonInterpreter        | 27 | kernel.BuiltComponentParams   | 47 |
| components.ether.ControlMessage      | 11 | components.py.StringList               | 27 | kernel.BuiltInType            | 64 |
| components.ether.DiscoveryConfig     | 11 | components.recorder.Configuration      | 28 | kernel.BusAddress             | 47 |
| components.ether.DiscoveryHubAddress | 8  | components.recorder.Recorder           | 28 | kernel.ClassTypeSpec          | 47 |
| components.ether.Endpoint            | 8  | components.recorder.RecorderState      | 28 | kernel.ClassTypeSpecV4        | 47 |
| components.ether.EnpointList         | 11 | components.recorder.RecordingSettings  | 28 | kernel.ClassTypeSpecV5        | 48 |
| components.ether.Ephemeral           | 8  |                                        |    | kernel.CompatibilityMode      | 60 |
| components.ether.Hello               | 8  |                                        |    | kernel.ComponentConfig        | 48 |
| components.ether.MaybeDeviceName     | 12 | components.recorder.RecordingSettings  |    | kernel.ComponentInfo          | 48 |
| components.ether.MaybeDiscoveryHub   |    | List                                   | 29 | kernel.ComponentState         | 60 |
| Port                                 |    | components.recorder.SelectionList      | 29 | kernel.CpuArch                | 60 |
| components.ether.MaybePortConfig     | 12 | components.replayer.Configuration      | 30 | kernel.CustomTypeData         | 64 |
| components.ether.MulticastAddress    |    | components.replayer.Replay             | 30 | kernel.CustomTypeDataV4       | 64 |
| Exclusions                           | 12 | components.replayer.ReplayStatus       | 31 | kernel.CustomTypeDataV5       | 64 |
| components.ether.MulticastAddress    |    | components.replayer.Replayer           | 30 | kernel.CustomTypeSpec         | 48 |
| Range                                | 9  | components.rest.Argument               | 32 | kernel.CustomTypeSpecList     | 66 |
| components.ether.MulticastDiscovery  | 9  | components.rest.Arguments              | 36 | kernel.CustomTypeSpecListV4   | 66 |
| components.ether.MulticastRange      | 12 | components.rest.Bus                    | 32 | kernel.CustomTypeSpecListV5   | 66 |
| components.ether.PinnedPort          | 9  | components.rest.BusNames               | 36 | kernel.CustomTypeSpecV4       | 49 |
| components.ether.PortBinding         | 11 | components.rest.BusSummary             | 32 | kernel.CustomTypeSpecV5       | 49 |
| components.ether.PortConfig          | 9  | components.rest.Buses                  | 36 | kernel.EnumTypeSpec           | 49 |
| components.ether.PortExclusions      | 12 | components.rest.ClientAuth             | 32 | kernel.EnumTypeSpecV4         | 49 |
| components.ether.PortRange           | 9  | components.rest.Configuration          | 33 | kernel.EnumTypeSpecV5         | 49 |
| components.ether.ProbePortRange      | 10 | components.rest.Error                  | 33 | kernel.EnumeratorSpec         | 49 |
| components.ether.ProtocolVersion     | 10 | components.rest.Events                 | 36 | kernel.EnumeratorSpecList     | 66 |
| components.ether.QueueConfig         | 10 | components.rest.HttpMethod             | 35 | kernel.EnumeratorSpecListV4   | 66 |
| components.ether.QueueEvictionPolicy | 11 | components.rest.Interest               | 33 | kernel.EnumeratorSpecListV5   | 67 |
| components.ether.Ready               | 10 | components.rest.InterestSummary        | 33 | kernel.EnumeratorSpecV4       | 50 |
| components.ether.SessionPresenceBeam |    | components.rest.InterestsSummary       | 36 | kernel.EnumeratorSpecV5       | 50 |
| 10                                   |    | components.rest.InvokeStatus           | 35 | kernel.EnvVar                 | 50 |
| components.ether.TcpDiscovery        | 10 | components.rest.Link                   | 33 | kernel.EnvironmentVarList     | 67 |
| components.explorer.Configuration    | 13 | components.rest.Links                  | 36 | kernel.EnvironmentVarList     | 67 |
| components.influx.Configuration      | 14 | components.rest.Notifcation            | 33 | kernel.ErrorCategory          | 60 |
| components.influx.Protocol           | 14 | components.rest.NotificationType       | 33 | kernel.ErrorData              | 50 |
| components.influx.SelectionList      | 14 | components.rest.Object                 | 35 | kernel.ErrorReport            | 50 |
| components.jsonrpc.AddedObjectEntry  | 19 | components.rest.ObjectEvent            | 34 | kernel.EventSpec              | 50 |
| components.jsonrpc.AddedObjectList   | 23 | components.rest.ObjectMethod           | 34 | kernel.EventSpecList          | 67 |
| components.jsonrpc.Configuration     | 20 | components.rest.ObjectSummary          | 34 | kernel.EventSpecListV4        | 67 |
| components.jsonrpc.ConnectionLimits  | 20 | components.rest.ObjectsSummary         | 36 | kernel.EventSpecListV5        | 67 |
| components.jsonrpc.Identity          | 20 | components.rest.Properties             | 37 | kernel.EventSpecV4            | 51 |
| components.jsonrpc.JsonRpcServer     | 15 | components.rest.RelType                | 35 | kernel.EventSpecV5            | 51 |
| components.jsonrpc.MaybeConnection   |    | components.rest.Session                | 34 | kernel.ExceptionKind          | 61 |
| Limits                               | 24 | components.rest.SessionSummary         | 34 | kernel.ExecError              | 51 |
| components.jsonrpc.MaybeControlBus   |    | components.rest.Sessions               | 37 | kernel.GitStatus              | 61 |
| Name                                 | 24 | components.rest.Status                 | 36 | kernel.IntegralType           | 61 |
| components.jsonrpc.MaybeFlag         | 24 | components.rest.SubscriptionOptions    | 34 | kernel.KernelApi              | 44 |
| components.jsonrpc.MaybeHigh         |    | components.rest.Subscriptions          | 34 | kernel.KernelParams           | 51 |
| BackpressureBytes                    | 24 | components.rest.Success                | 35 | kernel.LoadedComponentList    | 67 |
| components.jsonrpc.MaybeIdleTimeout  |    | components.rest.ThreadPoolSize         | 37 | kernel.LoadedComponentParams  | 52 |
| Seconds                              | 25 | components.rest.UpdateFrequency        | 37 | kernel.MaybeException         | 69 |
| components.jsonrpc.MaybeMax          |    | components.rest.UpdateTime             | 37 | kernel.MaybeF64               | 69 |
| BackpressureBytes                    | 25 | components.rest.Version                | 35 | kernel.MaybeSignal            | 69 |
| components.jsonrpc.MaybeMaxPayload   |    | components.shell.BufferStyle           | 41 | kernel.MaybeStackTrace        | 69 |
| Bytes                                | 25 | components.shell.CPrint                | 38 | kernel.MaybeTransportProtocol | 69 |
| components.jsonrpc.MaybeMember       |    | components.shell.ClearCurrentLine      | 39 | kernel.MaybeU64               | 70 |
| Selector                             | 25 | components.shell.ClearRemainingCurrent | 39 | kernel.MethodConstnessSpec    | 61 |
| components.jsonrpc.MaybeRateHz       | 25 | Line                                   | 39 | kernel.MethodSpec             | 52 |
| components.jsonrpc.MaybeStringList   | 25 | components.shell.ClearScreen           | 39 | kernel.MethodSpecList         | 67 |
| components.jsonrpc.MaybeSubscribe    |    | components.shell.Configuration         | 39 | kernel.MethodSpecListV4       | 67 |
| Block                                | 25 | components.shell.HideCursor            | 39 | kernel.MethodSpecListV5       | 67 |
| components.jsonrpc.MaybeTlsConfig    | 25 | components.shell.ListStyle             | 41 | kernel.MethodSpecV4           | 52 |
| components.jsonrpc.MaybeUpdateFreqHz |    | components.shell.MoveCursorAllLeft     | 39 | kernel.MethodSpecV5           | 52 |
| 25                                   |    | components.shell.MoveCursorDown        | 39 | kernel.NumericType            | 64 |
| components.jsonrpc.MemberSelector    | 23 | components.shell.MoveCursorLeft        | 39 | kernel.OpFinished             | 53 |
| components.jsonrpc.NamedSelection    | 20 | components.shell.MoveCursorRight       | 39 | kernel.OpNotFinished          | 53 |
| components.jsonrpc.ObjectInfo        | 20 | components.shell.MoveCursorUp          | 40 | kernel.OpState                | 65 |
| components.jsonrpc.ObjectInfos       | 23 | components.shell.NewLine               | 40 | kernel.OptionalTypeSpec       | 53 |
| components.jsonrpc.ObjectState       | 21 | components.shell.Print                 | 40 | kernel.OptionalTypeSpecV4     | 53 |
| components.jsonrpc.ObjectStateList   | 23 | components.shell.Query                 | 40 | kernel.OptionalTypeSpecV5     | 53 |
| components.jsonrpc.PropertyError     | 21 | components.shell.QueryList             | 42 | kernel.OsKind                 | 61 |
| components.jsonrpc.PropertyErrorList | 23 | components.shell.RestoreCursorPosition |    | kernel.PrecisionSleep         | 53 |
| components.jsonrpc.PropertyValueList | 24 |                                        |    | kernel.Priority               | 62 |
| components.jsonrpc.PropertyValuePair | 21 | components.shell.SaveCursorPosition    | 40 | kernel.ProcessData            | 53 |
| components.jsonrpc.RegisteredBundle  |    | components.shell.SetBgColor            | 40 | kernel.ProcessInfo            | 54 |
| Info                                 | 21 | components.shell.SetFgColor            | 40 |                               |    |

|                                   |    |
|-----------------------------------|----|
| kernel.SleepPolicy                | 65 |
| kernel.StringList                 | 68 |
| kernel.StructTypeFieldSpec        | 57 |
| kernel.StructTypeFieldSpecList    | 68 |
| kernel.StructTypeFieldSpecListV4  | 68 |
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| kernel.SystemSleep                | 58 |
| kernel.ThreadCreateErr            | 63 |
| kernel.TransportModeSpec          | 63 |
| kernel.U32List                    | 68 |
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| kernel.UnitCat                    | 63 |
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| kernel.VariantTypeFieldSpec       | 58 |
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| kernel.VariantTypeFieldSpecV4     | 58 |
| kernel.VariantTypeFieldSpecV5     | 58 |
| kernel.VariantTypeSpec            | 59 |
| kernel.VariantTypeSpecV4          | 59 |
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| kernel.VirtualClock               | 44 |
| kernel.VirtualKernelClock         | 44 |
| kernel.VirtualMasterClock         | 44 |
| kernel.WordSize                   | 63 |
| kernel.log.BasicFile              | 71 |
| kernel.log.ColorStderr            | 71 |
| kernel.log.ColorStdout            | 71 |
| kernel.log.Config                 | 71 |
| kernel.log.LogLevel               | 73 |
| kernel.log.LoggerConfig           | 71 |
| kernel.log.LoggerConfigList       | 73 |
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| kernel.log.SinkConfig             | 72 |
| kernel.log.SinkConfigList         | 73 |
| kernel.log.SinkNameList           | 73 |
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| kernel.log.Stderr                 | 72 |
| kernel.log.Stdout                 | 72 |
| kernel.log.Syslog                 | 72 |